

Survey paper

Robot learning in the era of foundation models: a survey

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ABSTRACT

The proliferation of large language models (LLMs) has fueled a shift in robot learning from automation towards general embodied artificial intelligence (AI). Adopting foundation models together with traditional learning methods for robot learning has increasingly gained interest in the research community and shown potential for real-life application. However, there is little literature that comprehensively reviews the relatively new technologies combined with robotics. The purpose of this review is to systematically assess the state-of-the-art foundation models in robot learning and to identify future potential areas. Specifically, we first summarized the technical evolution of robot learning and identified the necessary preliminary preparations for foundation models, including the simulators, datasets, and foundation model framework. In addition, we focused on the following four mainstream areas of robot learning, including manipulation, navigation, task planning, and reasoning, and demonstrated how the foundation model can be adopted in the above scenarios. Furthermore, critical issues that are neglected in the current literature, including robot hardware and software decoupling, dynamic data, generalization performance in the presence of humans, etc., were discussed. This review highlights the state-of-the-art progress of foundation models in robot learning. Future research should focus on multimodal interaction, especially dynamics data, robotics-specific foundation models, AI alignment, etc.

1. Introduction

Robots have played an important role in various scenarios, including industrial [1], medical [2], service [3], and special robot [4] industries. With the increasing task complexity and working environment variability, the demands for robot tasks have shifted from fixed automation to general artificial intelligence, where robot learning will be the core enabling technique of autonomous systems [5].

The field of robot learning lies at the junction of machine learning and robotics. Its primary focus is enabling robots to acquire new skills or adapt to their environment by utilizing learning algorithms. Some examples of skills that learning algorithms target include sensorimotor skills (like locomotion, grasping, and active object classification), interaction skills (such as co-manipulating objects with humans), and linguistic skills (including the grounded and situated meaning of human language). Learning can be acquired through autonomous self-exploration or demonstration-based teaching.

Extensive research studies have been performed on how to realize robot learning. Traditional robot learning techniques are generally divided into imitation learning [6–9] and reinforcement learning [10, 11]. Imitation learning enables robots to learn new skills by mimicking human behavior, while reinforcement learning allows robots to optimize the outcome of skill execution. Imitation learning is an important way to initialize and improve learning efficiency in reinforcement learning. However, they all suffer from certain limitations. Imitation learning is a straightforward and stable form of supervised learning. It requires labeled behavior data and is unable to surpass human-level performance. Although reinforcement learning has the potential to surpass human-level performance, it requires defining the reward function and addressing the policy exploration challenge. It may also encounter convergence issues and instability during training. The characteristics of robot learning are that robots are physically embodied and environmentally situated. Thus, a critical challenge in robot learning is to close the perception-action loop in the practical scenarios, and there are still

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some problems in the practical application of traditional robot learning, including insufficient generalization of tasks, insufficient environmental adaptability, low execution accuracy, and lack of planning and reasoning capabilities [12–14].

The foundation model concept [15] was first introduced by the Center for Research on Foundation Models (CRFM) at Stanford's Institute for Human-Centered Artificial Intelligence (HAI). Foundation models, including GPT-n, BERT, and DALL-E, are trained on a broad spectrum of data, typically using large-scale self-supervision, and can be adapted for a variety of downstream tasks. Characterized by scaling law, emergence [16], and generality [15], foundation models represent a significant paradigm shift in artificial intelligence. Consequently, numerous AI systems across various fields are either being built upon or heavily incorporating foundation models. Recently, with the release of ChatGPT, foundation models, particularly multimodal foundation models, presented both opportunities and challenges for robot learning. Foundation models have demonstrated significant potential in achieving human-level intelligence, thereby leading to a surge of research in robotics based on foundation models. Leveraging multi-domain prior knowledge, foundation models have made breakthroughs in understanding complex tasks, engaging in continuous dialogue, and performing zero-shot reasoning. However, foundation models still lack the ability to perceive the external environment, which can be addressed by employing the multi-modal foundation model incorporating 2D&3D vision, LiDAR, voice, inertial motion unit (IMU), etc. Researchers have developed task-specific robot learning architectures to fully unleash the potential of the foundation models and address the current challenges in robot learning so that robots can learn human behavior and skillfully undertake a series of tasks. However, these models were proposed independently and relatively recently. Upon thorough investigation, existing surveys on robot learning are primarily focused on a single task and predominantly rely on traditional methods [5,17–21]. There are few literature reviews on robot learning based on foundation models [22–31]. Therefore, it is crucial to construct an overall summary analysis of existing robot learning research based on foundation models, which is of great significance for a comprehensive understanding of the field and to provide inspiration for future research [32].

We organized this survey according to the solutions to the following three research questions. What platforms does the embodied AI need? What foundation model algorithms, strategies, and mechanisms are

currently used for downstream tasks in robot learning? What are promising future research areas in robot learning for current research questions? In this paper, we systematically survey the relatively new field of foundation-model-based robot learning and establish a clear taxonomy for the existing research in the field, focusing on the four aspects of robot learning: manipulation, navigation, task planning, and reasoning. We also identify several challenges in this field and discuss potential future directions.

The overall framework (Fig. 1) is organized as follows: historical development leads to the necessity of applying foundation models to robot learning. The challenges of the new era lead to the necessary tools. Downstream tasks are divided into four categories, each of which plays a corresponding role. Manipulation is the basis for robots to perform specific tasks. Navigation provides support for position movement for manipulation, ensuring that the robot can reach the location where manipulation is required. Task planning is responsible for designing the overall execution strategy of the task and coordinating the order and logic of manipulation and navigation. Reasoning ensures that manipulation, navigation, and task planning can adapt to environmental changes and successfully complete the task through real-time analysis and adjustment. The four work together to enable robots to efficiently complete tasks in complex and changing environments. Finally, we discuss the existing problems and subsequent development directions.

We select references based on the following criteria: 1) Our primary objective is to encompass all significant milestones in robot learning during the advent of foundation models across diverse tasks. 2) the diversity and significance of subfields and research groups should be considered. 3) A valuable source entails conducting a reverse search beginning with highly influential publications [33].

We conducted a survey of existing similar literature. Before we published it on arXiv, there were few relevant reviews. Before that, large language models mainly focused on the text field. We focus on the progress of foundation models in the field of physics, especially the combination with robots. We are one of the first to publish a review of the research combining the foundation models with robots. And some studies followed our work [24,29,30]. In addition, compared with other similar review papers by Sep. 2024, our review paper exhibits several distinctive advantages compared to other existing reviews. Compared with [34], [26], and [25], we have a fine-grained classification of foundation models. The coverage of the models, such as LLM, VLM,

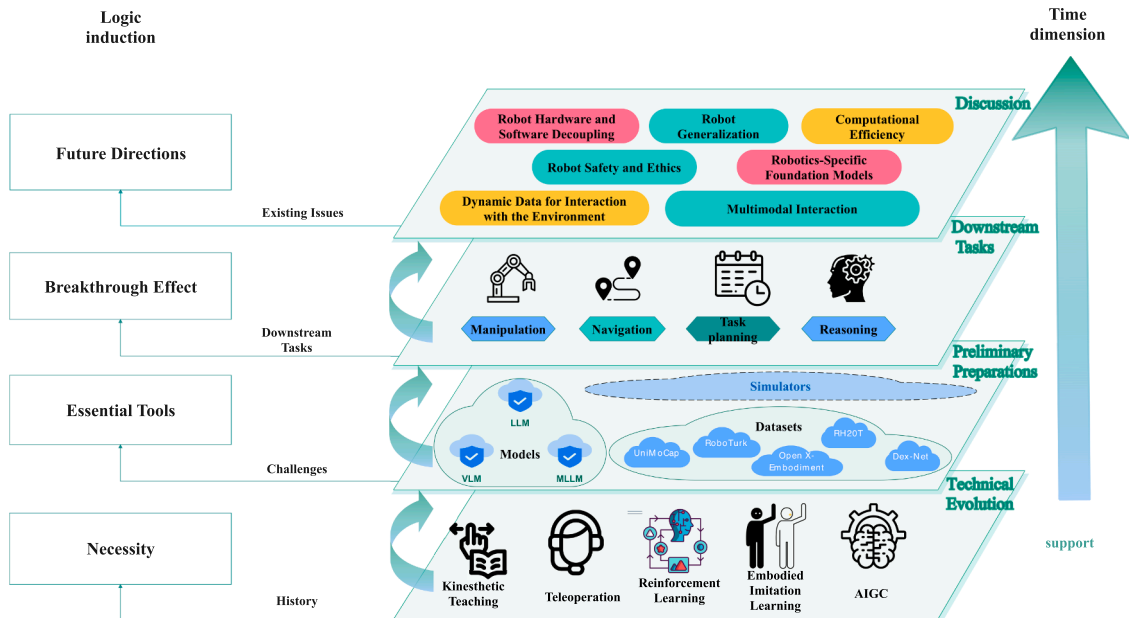


Fig. 1. Overall structure of this survey paper.

MLLM, and diffusion models, is wider than that of others. During the comparison of [24,29] and [28,30], we thoroughly examined the historical categorization of robot learning techniques and conducted a detailed analysis of the robot datasets and simulators. Simultaneously, we prioritized the comparison and analysis of how foundation models are applied to various downstream tasks. This holistic approach ensures a more detailed and multifaceted understanding of the current landscape and considers the critical need for robotic foundation models, offering valuable insights that are not as extensively covered in other surveys.

The contributions of this paper can be highlighted in the following aspects:

- (1) This review is the one of the pioneer works that focuses on combining foundation models with robots and systematically categorizes the preliminary tools and methods in the presence of the new challenges of foundation models.
- (2) In the domain of integrating foundation models with robots, four primary scientific issues have been identified: robot manipulation, navigation, task planning, and reasoning. A list of common applications and scenarios where foundation models can be effectively combined with robots to address various difficulties has been compiled. A comparative analysis was conducted on the integration of foundation models with robots to address various challenges across different problem domains.
- (3) Critical bottlenecks in the existing problems were identified, and both possible solutions to these bottlenecks and future development directions were discussed in depth.

2. Overview

2.1. Technical evolution

Artificial Intelligence Generated Content (AIGC) is a new content creation method. AIGC generates content based on artificial intelligence technology to meet user needs. It is considered a technology with broad application prospects that complements traditional content creation methods (professionally generated content (PGC) and user-generated content (UGC), etc.) [35]. The exploration and discovery of robot learning have gone through different stages of historical evolution (Fig. 2). They can be summarized as the following four categories: robot learning through teaching programming (TP), including kinesthetic teaching and teleoperation, reinforcement learning (RL) in interaction, embodied imitation learning (IL) [36,37] and learning in AIGC-based generative models [38–40].

Kinesthetic teaching [45] involves physically maneuvering the robot to perform the necessary actions while the robot's onboard sensors record status information, which is then used to generate training data for the machine learning model. Although this method is relatively simple, the quality of the demonstration relies heavily on the operator's ability to perform movements with flexibility and smoothness. While it proves highly effective for robotic arms, its application becomes more challenging on other platforms, such as legged robots or dexterous hands. Another demonstration method is teleoperation [46], which enables trajectory learning, task learning, grabbing, or more advanced tasks by providing external input to the robot through handles, graphical interfaces, or other methods. There are currently a variety of interactive devices (such as haptic devices or VR interactive devices). In contrast to kinesthetic teaching, teleoperation allows for remote implementation, eliminating the need for the user and the robot to be physically present

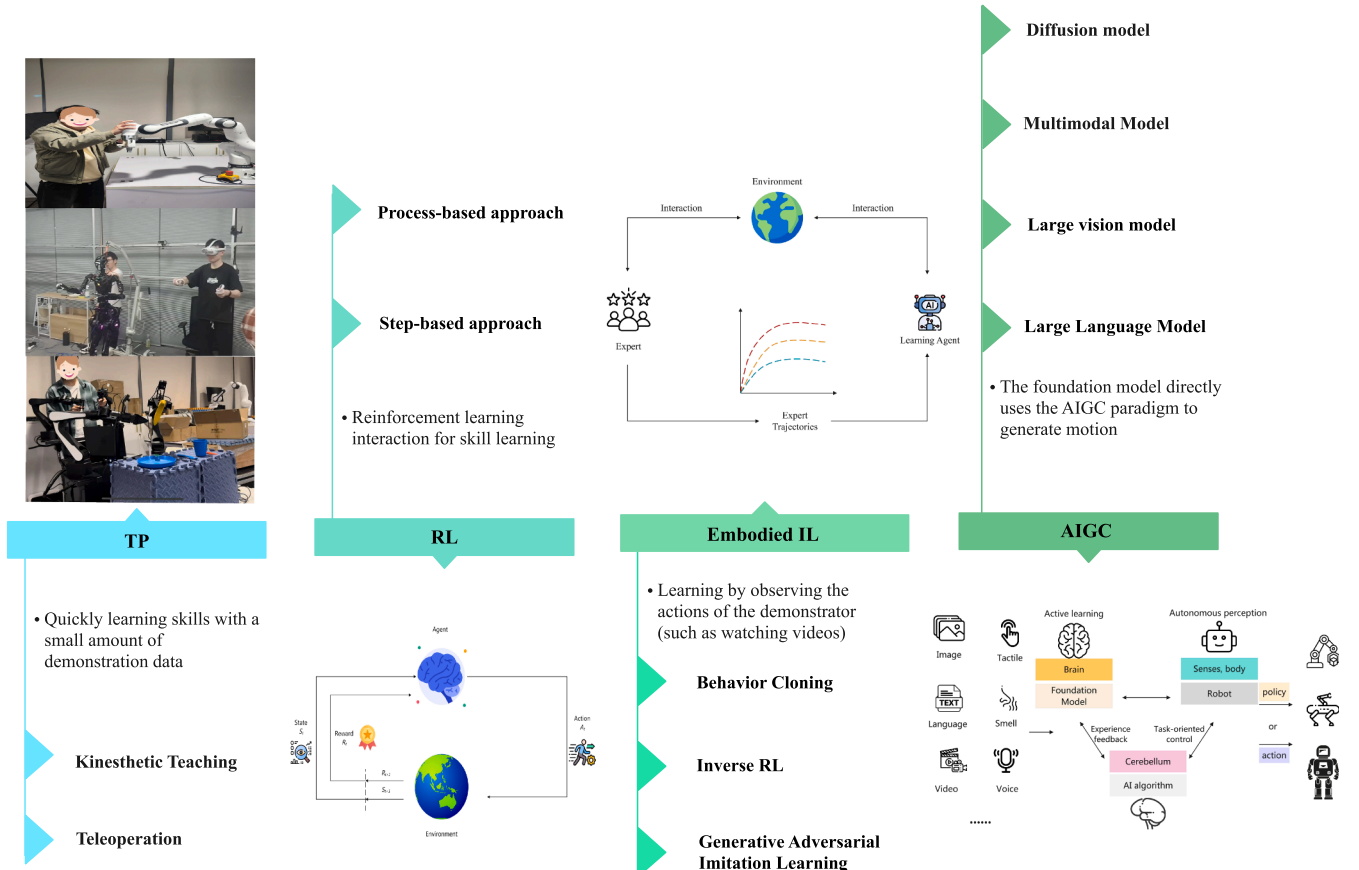


Fig. 2. Technical evolution [41–44].

at the same site. Limitations of teleoperation include the additional work required to develop input interfaces, a longer user training process, and usability risks posed by external devices.

Reinforcement learning involves determining optimal actions in a given situation to maximize the digital revenue signal. Rather than being explicitly instructed on which actions to take, learners must independently identify the actions that yield the greatest benefits. Reinforcement learning focuses on finding a balance between exploring unknown territory and leveraging current knowledge. A typical framework for a reinforcement learning scenario: an agent takes an action in the environment, and the action is interpreted as a reward and state representation, which is then fed back to the agent [47].

Embodied imitation learning [48] mainly involves robots learning by observing the actions of the demonstrator (such as watching videos). In this method, the presenter utilizes their own body to perform the task while an external device records their movements. This approach proves to be the most straightforward for presenters, as it does not necessitate any training in the presentation process. Furthermore, this approach can be extended to robots with multiple degrees of freedom, including non-anthropomorphic robots. However, enabling this approach involves mapping human actions to robot-executable actions, which presents challenges such as occlusion, rapid movement, and sensor noise during demonstrations.

By integrating foundation models [39] with robots, based on their embodied intelligence, zero-shot training is achieved for motion planning and execution in real-world scenarios, resulting in human-like motion control logic and capabilities. This type of robot obtains general knowledge from foundation models and addresses cognitive deficiencies through perceptual intelligence, such as vision. This marks the initial steps towards the industrialization of embodied intelligent robots.

2.2. Preliminary preparations

Developing a learning model for robots is not an easy task [49,50], given the challenges of technical issues [51,52] including computing resources, algorithms and data. A feasible approach is to perform incremental development or experimental verification on the basis of existing models. In this section, we briefly organize publicly available resources [53–59] for robot learning, including simulators, datasets [60–63], and foundation models [50].

As developing and testing applications with real robots is expensive and time-consuming, simulation has emerged as a crucial component in the field of robotics application development. The validation of applications in simulation prior to deployment on robots can shorten iteration time by identifying potential issues at an early stage. Simulation facilitates the testing of corner cases or scenarios that could pose risks in the real world. To validate and assess the performance and effectiveness of robot learning, it is beneficial to conduct initial testing in a simulator before transitioning to the real-world environment. Several simulation frameworks [64] currently exist for training in robot learning simulations and are summarized in Table 1.

To validate the correlation performance in the simulation experiment, it is essential to have a dataset. There are two types of datasets: static data and dynamic data. Static data refers to related datasets collected on the Internet, and dynamic data refers to data generated through interactions with robots in a real-world environment. Currently, the majority of existing datasets consist of static data, and the specifics of prominent datasets are summarized in Table 2. Obtaining dynamic data is challenging, and it is relatively scarce.

Foundation models [100] are inseparable from the support of computing power, algorithms, and data. Computing power is determined by hardware components such as the CPU, GPU, and memory [101]. Higher-performance chips are necessary for the training and construction of the neural network in foundation models. Research institutes currently employ diverse algorithms to implement foundation models. However, the primary difficulty lies in acquiring high-quality

data. High-quality data plays a crucial role in facilitating AI training and tuning. The following is a summary of notable foundation models in Table 3.

Foundation models encompass several categories:

1. Large Language Models: These include models such as BERT, GPT-3, and PaLM, primarily used in natural language processing tasks.
2. Large Vision Models: This category consists of Visual Transformer Models and Visual Generative Models. Examples of Visual Transformer Models include ViT and Swin Transformer, which are used in computer vision tasks. Visual Generative Models, such as diffusion models and GANs, are employed to generate visual content.
3. Large Multimodal Models: Models like CLIP [102] and ALIGN [103] integrate vision and language by utilizing cross-modal contrastive learning.
4. Large Embodied Models: Large embodied models, such as PaLM-E, are generally used for decision-making or task execution.

In addition to LLM methods, large vision models play a role in the field of robotics. For example, the application of diffusion models can help improve strategy representation capabilities and learn diverse task skills. In the field of robot control, research has found that diffusion models can predict closed-loop action sequences while maintaining temporal stability. For example, the core component of the DALL-E-Bot method is the large-scale pre-trained image diffusion model DALL-E2. DALL-E-Bot uses a modular approach to arrange targets like humans. First, the initial observation image is converted into an instance-level object description consisting of segmentation masks, object text descriptions, and CLIP visual feature vectors. Next, the DALL-E-Bot begins building textual prompts describing objects in the scene and passes these into the DALL-E model to create target images for the rearrangement task. Then, CLIP visual features are used to match objects in the initial and generated images, and their pose information is estimated by aligning their segmentation masks. Finally, the robot rearranges the targets in the scene based on the estimated pose information.

What's more, there are also large multimodal models combined with robotic technology to produce better results. Vision-language models (VLMs), such as CLIP, ALIGN, achieve the alignment of vision and language through cross-modal contrastive learning. Based on the open source large multimodal model OpenFlamingo, the open source and easy-to-use RoboFlamingo robot operation model has been developed, which can be trained with only a single machine. With simple and small amounts of fine-tuning, VLM can be turned into Robotics VLM, making it suitable for language-interactive robot operation tasks.

Large Embodied Models: Embodied intelligence aims to enable robots to interact with the environment and complete tasks in a more autonomous and intelligent way by combining intelligent algorithms with the perception, action, and environment interaction of physical entities. In response to challenges such as low success rate of instruction decomposition, poor completion of long horizon tasks, planning that does not consider environmental visual information, and poor adaptability to new tasks, there has been a lot of progress in large embodied models (Code as Policies, Palm-E, SayCan, ViLa, etc.). In addition, many new works have emerged in cutting-edge applications such as agent learning tools, multi-agent collaboration, multi-modal human-computer interaction, and 3D world models. One type is a large decision-making model [149]. Currently, there are two main paradigms of large decision-making models. One paradigm is based on large language models and is coordinated with API selection decisions. The other is centered on a large reinforcement learning model to perform low-level decision-making control. There is no specific API that can exactly solve the low-level decision-making, and reinforcement learning strategies need to be used for direct generalization. Large reinforcement learning models include large policy models (e.g., Gato, DB1), large value models (e.g., Q-Transformer), large environment models (e.g., TDM, Genie), and large multi-agent models (e.g., MADT, MAT).

Table 1
Simulation framework.

	MineDojo [65]	Habitat3.0 [66,67]	BEHAVIOR-1K [68,69]	iGibson 1.0 [70]	AI2-THOR 2.0 [71]	BabyAI [72]	PyBullet	PyRobot [73]	Isaac Sim [74]	RFUniverse [75]	Unisim [76]
Simulator	MineDojo	Habitat Sim	OMNIGIBSON	iGibson	AI2-THOR 2.0	MiniGrid	PyBullet	Gazebo	Omniverse	RFUniverse	Unisim
Dataset	MineDojo	Habitat Synthetic Scenes Dataset	BEHAVIOR-1K	iGibson dataset	iTHOR, RoboTHOR, ProcTHOR-10K, ArchitecTHOR	-	-	-	-	-	-
Supported Task	Minecraft Game	Navigation, Manipulation	Household Activities	Navigation, Manipulation	Navigation, Manipulation	Navigation, Manipulation	Manipulation, Locomotion, Control	Navigation, Manipulation	Navigation, Manipulation	Complex Dynamics Tasks	Self- Driving
Sensors/ Sensor Signals	Video	RGB-D Cameras, GPS	On-board virtual sensors (RGB, Depth images, LiDAR)	RGB images, Rendering of normals, Depth, Point clouds, Virtual LiDAR signals, and Optical scene flow	Cameras and the environment metadata	-	-	Camera, Distance, Proximity sensors, Laser, Force sensors	RGB-D, Lidar, and IMU	Vision, IR, DIGT	Cameras, LiDAR
Language	JAVA, Python	C+ +, Python	Python	Python, C	C#, Python	Python	Python	C+ +, Python	Python	C#	-
Supported OS	Linux, Mac OS, Windows	Linux, Mac OS, Windows	Windows, Ubuntu	Windows, Linux, Mac	Mac OS, Ubuntu	Windows, Linux, Mac	Windows, Linux, Mac OS	GNU, Linux (Ubuntu)	Windows, Linux	Windows, Linux	-
Scenes	-	211	50	15	120	19	9	-	-	-	103
Objects	-	18k	5000+	570	84	-	-	-	-	-	-
Physics Engine	Minecraft game	Bullet	PhysX 5	PyBullet	Unity	-	Bullet Physics SDK	ODE, Bullet, Simbody, DART	PhysX	Unity	-
3D Rendering Engine	Minecraft game	Magnum	Nvidia Omniverse	Physics-Based Rendering	Unity	-	OpenGL	OGRE	RTX	Unity	-
Speed	-	140–250 steps/ s	-	100 steps/s	90–180 steps/s	-	240 steps/s	-	-	-	-
Robot Family	-	Spot Robot	-	Articulated Robots, Fetch Robot	-	-	R2d2 Robot	Mobile, Humanoid, Industrial	Quadruped robots, Franka robot, UR, and wheeled robots	-	-
Supported Tools	-	-	-	Grippers, Locobot	-	-	-	Grippers	Manipulators	-	-
Computing Resource	8 × V100 GPUs	-	-	-	-	20–50GPU	A high-end desktop GPU	-	-	-	-
Open Source	Open	Open	Open	Open	Open	Open	Open	Open	No	Open	No

Table 2
Comprehensive robot datasets.

	Data Type	Data Volume	Task	Object numbers	Scenes	Skills
Open X-Embodiment [77]	Demonstrations	4435.41 GB	Manipulation	-	-	527
HoloAssist [78]	RGB, depth, head pose, 3D hand pose, eye gaze, audio, IMU, and text	166 h	Collaboratively manipulation	-	-	20
UniMoCap	Text-motion mocap	34 k motions and 66 k annotations	Describing the actions being performed in mocap sequences	-	-	-
ARMBench [79]	Images and videos	235 k	Object segmentation, Object Identification, and Defect Detection	190 k	1	-
RT-1 [80]	Instruction and image tokenization	130 k	Manipulation	-	3	744
RoboTurk [81]	Demonstrations	137 h	Block Lifting (lifting), Bin Picking (picking), and Nut-and-peg Assembly (assembly)	-	2	-
Raven [82]	Images and RPM	1120000 images and 70000 RPM	Reasoning, VQA	-	-	-
RoboNet [83]	Video frames	15M video frames	Manipulation	-	4	-
GSO [84]	3D images	13 G	Manipulation, Navigation, etc.	1030	-	-
Meta-world [85]	Video, trajectory	-	Manipulation	-	50	50
RLBench [86]	RGB, depth, and segmentation masks	-	Manipulation	-	-	100
M2DGR [87]	RGB image, 3D point cloud, inertial data, GNSS signals	36 sequences (about 1 TB)	SLAM	-	-	-
OBJECTFOLDER 2.0 [88]	Object files (containing the complete multisensory profile)	1000 object files	Object scale estimation, Contact localization, and Shape reconstruction	1000	-	-
Google Brain Robot Data [89]	Images	~ 800k grasp attempts	Manipulation	-	-	-
Dex-Net 2.0 [90]	Point cloud	6.7 M point clouds	Manipulation	1500	-	-
Bridge Data [91]	Demonstrations (video)	7200 demonstrations	Household kitchen tasks	-	10	71
RH20T [92]	RGB image, Depth image, Binocular IR images, Robot joint angle, Robot joint torque, Gripper Cartesian pose, Gripper width, 6-DoF Force/Torque, Fingertip tactile	~ 110 k	Learning task and motion planning	-	-	147
Radish	Odometry, laser and sonar data, sensor data, environment maps	Multi datasets (Use Sal200 Synthetic, Robonaut Sensor)	Robotics dataset community	-	-	-
Daily Interactive Manipulation [93]	Position, orientation, force, and torque of objects	3199 trails	Manipulation	-	-	33
Robot @ Home [94]	Intensity images, depth images, and 3D point clouds, laser scanner data, topological information	9.6 G	Object/room instance recognition, object segmentation, data compression/transmission	-	-	-
TEAch [95]	Language	3047 sessions	Navigation, Dialogue	-	-	-
Robotic 3D Scan Repository	3D point clouds	Multi datasets (FHG Campus, FireAcademy)	SLAM (navigation)	-	-	-
MRPT	Sensors from 2D laser scanners up to RTK GPS, stereo cameras or 3D ToF cameras	Multi datasets (Kenmore, Edmonton 2002)	Mobile robotics and computer vision	-	-	-
ImageNet [96]	Image	1.4M+ images	Computer vision	-	-	-
EgoNet [96]	Image	1.5M video frames	Manipulation	-	100+	-
OakInk-Image [97]	Image	230k image frames	Hand Mesh Recovery and Hand-Object Pose Estimation	100	-	-
OakInk-Shape [97]	Obj models	62k hand-object poses and models	Grasp Generation, Intent-based Interaction Generation, and Handover Generation	-	-	-
HANDAL [62]	Image frames	306k image frames	Manipulation	210	-	-
ScanScribe [98]	3D scan and text	2995 RGB-D scans	3D vision-language grounding	56.1k	1185	-
Sound-Action-Vision Dataset [99]	Sound, RGBD, tracking location	15000 interactions	Interplay of action and sound	60	-	-

Due to the substantial cost of model pre-training, utilizing public API [150,151] allows for remote execution of inference tasks.

3. Downstream tasks

In order to facilitate the interaction and collaboration between robots and humans, we hope that robots have human-like intelligence. We selected the following four mainstream tasks based on Brodmann areas of the cerebral cortex [152]. Therefore, this paper focuses on the four common functions of combining the foundation model and the robot (i. e., the four downstream tasks proposed in the paper, namely manipulation, navigation, task planning, and reasoning) [27]. The progress of these downstream tasks can bring great convenience to people's lives,

especially in dangerous scenes and heavy labor scenes where robots replace humans, so the breakthrough progress of downstream tasks has great social value.

According to the hierarchical idea, task planning and reasoning focus on the top level, which is similar to the function of the brain, while manipulation and navigation focus on the bottom level, which is similar to the function of the cerebellum [153,154]. Specifically, from top to bottom, reasoning focuses on "why do this" or "how to make decisions in complex situations" (understanding the environment and tasks, and making intelligent judgments). Task planning focuses on "how to do it" or "how to arrange tasks" (task decomposition). Navigation focuses on "where to go" (positioning and path selection). Manipulation focuses on "what to do" (executing actions). Under this classification method, task

Table 3
Foundation models.

	Foundational Model	Data Type	Data Size	Parameters Scale	Tokens Scale	GPU Numbers	Training Time	Publisher	Open Source
Language Models	ChatGLM [104]	Text	1.2 T English, 1.25 T Chinese (130 B)	6 B–130 B	400 B–1 T	96 DGX-A100 GPU (8 ×40 G)	60 days	Tsinghua	✓
	T5 [105]	Text	-	60M–11 B	1 T	-	-	Google	✓
	GPT3 [106]	Text	-	175 B	300 B	-	-	OpenAI	×
	LaMDA [107]	Text	1.56 T words	2B–137 B	2.81 T	1024 TPU-v3 chips	57.7 days	Google	×
	LLaMA [108]	Text	4.7 T	7B–65 B	1.4 T	2048 A100 GPU	21 days	Meta	✓
	MOSS [109]	Text	700 B words	16 B	-	-	-	Fudan	✓
	InternLM [110]	Text	1.6 T	7B, 20 B, 104 B	1.6 T	-	-	University Shanghai AI Laboratory, SenseTime	✓
	Baichuan 2 [111]	Text	127 M vocab size	7B, 13 B	2.6 T	8 A800 GPUs	-	Baichuan Inc.	✓
	Qwen [112]	Text	3 T	7 B, 14 B	3 T	-	-	Alibaba	✓
	PANGU-Σ [113]	Text	2158 GB	1.085 T	329B	512 Ascend 910 NPU	100 days	Huawei	×
	ColossalChat [114]	Text	104k Chinese and English	7 B	54 M	8 A100-40G GPUs	-	HPC-AI Tech	✓
	Alpaca [115]	Text	52k instructions	7 B	-	8 80 GB A100s	3 h	Stanford University	✓
Vision Models	SAM [116,117]	Image	11 M images, 1.1 B masks	91 M-636 M	-	256 GPUs	68 h	Meta	✓
	DINOv2 [118]	Image	1.2 B images	~1 B	-	8 V100–32 GB GPUs	2 days	Meta	✓
	ViT [119]	Image	317 M images	86 M	-	-	-	Google	✓
	VideoMAE V2 [120]	Video	1.35 M clips	1 B	-	64 A100 GPUs	14 days	Nanjing University	✓
Multilingual Models	BLOOM [121]	Text	341 B tokens of text	176 B	366B	48 nodes with 8 NVIDIA A100 80 GB GPUs	3.5 months	Google	✓
	Vicuna [122]	Text	70k samples	13 B	-	8 A100 GPUs	1 day	UC Berkeley, etc.	✓
Multimodal Models	Claude	-	-	-	-	-	-	Anthropic	×
	GPT-4 [123]	Text, Image	-	-	-	-	-	OpenAI	×
	GPT-4V [124]	Text, Image	-	-	-	-	-	OpenAI	×
	MiniGPT-5 [125]	Text, Image	-	7 B	-	4 A6000 GPUs	-	University of California, Santa Cruz	✓
	ERNIE Bot [126]	Text, Image	4 TB Chinese text corpora	100 B–260 B	-	-	-	Baidu	×
	CLIP [102]	Text, Image	400 M (image, text) pairs	~ 1.6 B	-	-	-	OpenAI	✓
	DALL-E [127]	Text, Image	250 M text-images	12 B	-	-	-	OpenAI	✓
	VILT [128]	Text, Image	4 M images and 9.8 M captions	87.4 M	-	64 NVIDIA V100 GPUs	-	NAVER AI Lab	✓
	VisionLLM [129]	Text, Image	238k images	7 B	-	4 × 8 NVIDIA A100	-	Shanghai AI Laboratory	✓
	Prismer [130]	Text, Image	11 M images, 12.7 M (images, text)	275 M–1.6 B	-	-	-	Imperial College London, etc.	✓
	BEiT-3 [131]	Text, Image	15 M images and 21 M image-text pairs	1.9 B	-	-	-	Microsoft	✓
	LLaVA [132]	Text, Image	~ 80k images	13 B	-	-	-	University of Wisconsin-Madison	✓
	USM [133]	Text, Audio	12 M hours of speech, 28 B sentences of text	2 B	-	-	-	Google	×
	AudioPaLM [134]	Text, Audio	-	600 M, 2 B	-	-	-	Google	×
	Whisper [135]	Text, Audio	680 M hours of audio	39 M–1550 M	-	-	-	OpenAI	✓
	MMS [136]	Text, Audio	107.4 h	0.3 B, 1 B	-	48(0.3B) or 64 (1B) A100 GPU	2.3 h(0.3B) or 3.5 h (1B)	Meta	✓
	DALL-E3 [137]	Text, Image	1 B images	-	-	-	-	OpenAI	×
	Meta-transformer [138]	12 Types (Text, Images, Point Clouds,	Total ~ 449 M	85 M, 302 M	-	-	-	Chinese University of Hong Kong, etc.	✓

(continued on next page)

Table 3 (continued)

	Foundational Model	Data Type	Data Size	Parameters Scale	Tokens Scale	GPU Numbers	Training Time	Publisher	Open Source
Embodied Agents		Audio, Video, etc.)							
	Palm-E [139]	Text, Image	-	12 B, 84 B, 562 B	-	-	-	Google	×
	VIMA [140]	Text, Image, Video	650k trajectories	2 M–200 M	-	8 NVIDIA V100 GPUs	1 days	Stanford University	✓
	RT-2 [141]	Text, Image, Action	10 B image-text pairs	5 B, 55 B	-	-	-	Google DeepMind	×
	RT-X [77]	Text, Image, Action	-	35 M, 5 B, 55 B	-	-	-	Google DeepMind	✓
	LM-Nav [142]	Text, Image, Action	-	(Based on GPT3, ViNG, CLIP)	-	-	-	UC Berkeley	✓
	VoxPoser [143]	Text, Image	-	(Based on GPT4)	-	-	-	Stanford University	✓
	VOYAGER [144]	Text, Image	-	(Based on GPT4)	-	-	-	NVIDIA	✓
	DIAL [145]	Text, Image	80 k trajectories	(Based on CLIP)	-	-	-	Google	×
	Gato [146]	Text, Image, etc.	63 M Episodes	1.2 B	1.5 T	-	-	DeepMind	×
	RoboCat [147]	Text, Image, etc.	-	1.2 B	-	-	-	DeepMind	✓
	MOMA-Force [148]	Vision, Force	-	-	-	-	-	ByteDance	×

planning and reasoning are more involved in decision-making and strategy, while manipulation and navigation are more inclined to the physical execution level.

The target and output of robot learning is to help the robot tactfully fulfill a certain task via acquiring a behavior or skill. Common target

tasks include robot manipulation, navigation, task planning, and reasoning (Fig. 3). This section investigates and reviews how different works combine foundation models for different downstream tasks to achieve robot learning goals [16,155].

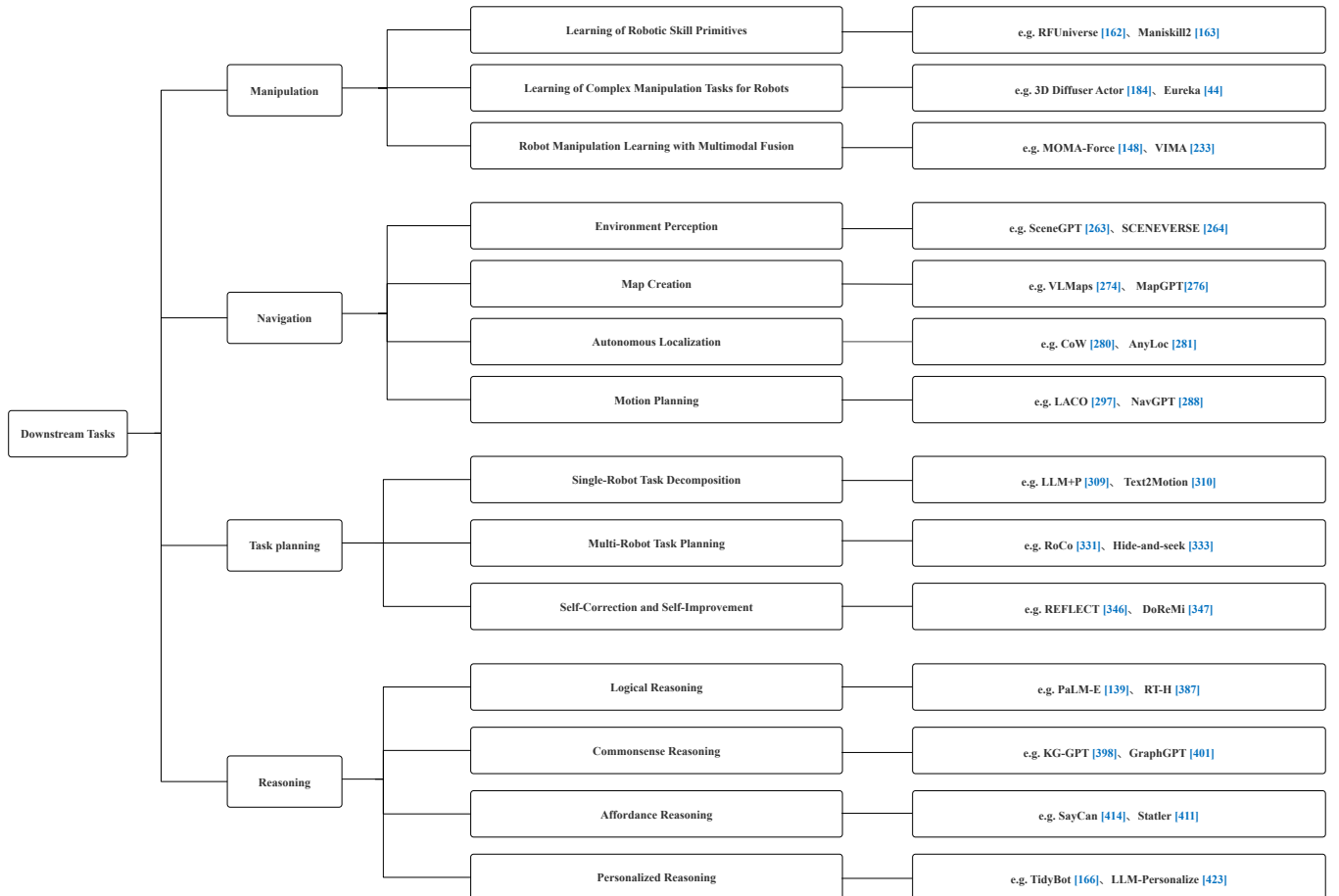


Fig. 3. Downstream tasks for robot learning.

3.1. Manipulation

3.1.1. Characteristics and challenges

Manipulation refers to the robot's physical behavior execution, including grasping, moving, rotating, etc. The core function is to convert control signals into specific physical actions. Tasks, despite being commonly perceived as simple and routine, such as washing dishes, cutting vegetables, and packing clothes, are still challenging for robots and are at the forefront of robotics research [156]. The robot manipulation problem deals with how a robot learns to manipulate its surrounding environment [5]. It can be formulated as the ability to connect a starting state to a goal state through successive actions, during which the manipulation task is represented by a set of points denoting the starting and goal states, along with the constraints imposed on the transition states [157]. Furthermore, the ability to make physical contact is critical to the development of manipulation skills. Currently, robots can only effectively grab and release certain types of objects and perform a variety of simple manipulation actions such as throwing, sliding, pushing, and poking. Challenges emerge when these actions need to be performed in cluttered environments or when more complex interactions are necessary [14]. Foundation models have the capability to engage in interactive dialogues with users and receive and process various types of data, including images, text, and speech. They can utilize multimodal information, such as vision, to guide the actions of the robot or generate corresponding code or action sequences. These capabilities enable robots to effectively adapt to intricate and ever-changing environments, eliminating the need for defining every single possible scenario in advance [155]. In this section, we survey the recent research results and the latest progress of foundation models in the direction of robot manipulation (Fig. 4), including learning robotic skill primitives, learning complex robot manipulation tasks, and robot

manipulation learning with multimodal fusion.

The scope of manipulation discussed in this article focuses on the following. Skill action primitives focus on the simple level. The action primitives are the smallest units that constitute all manipulation. Action primitive learning only learns how to perform a single action, and the operation level is low, and does not involve how to perform complex tasks. Complex tasks focus on the combination level of action primitives. Complex tasks are action flows formed by combining action primitives in time and space dimensions. Complex task learning focuses on combining multiple action primitives to form a complete task flow, rather than simply executing an action. Multimodal learning optimizes the robot's action decisions through different sensor inputs, focusing on the closed loop between perception and execution, and does not directly participate in the execution of tasks or actions. Multimodal perception and interaction provide the necessary information and feedback support, enabling robots to make accurate decisions and dynamic adjustments when performing complex tasks to ensure the smooth completion of tasks. We can use these three different levels to evaluate the intelligence and reliability of robot execution.

3.1.2. Methods

3.1.2.1. Learning of Robotic Skill Primitives. Action primitives are the basic building blocks of robot skills and constitute the smallest unit of all manipulation. Skills are the flow of actions of some basic functions in the time dimension. The approach involves breaking down the problem of robot skill learning into several easily comprehensible skill primitives or movement primitives. Subsequently, suitable and efficient learning methods are devised for these skill primitives, allowing the robot to acquire manipulation skills by combining these primitives. This enables the robot to adapt to new environments and generalize its abilities to

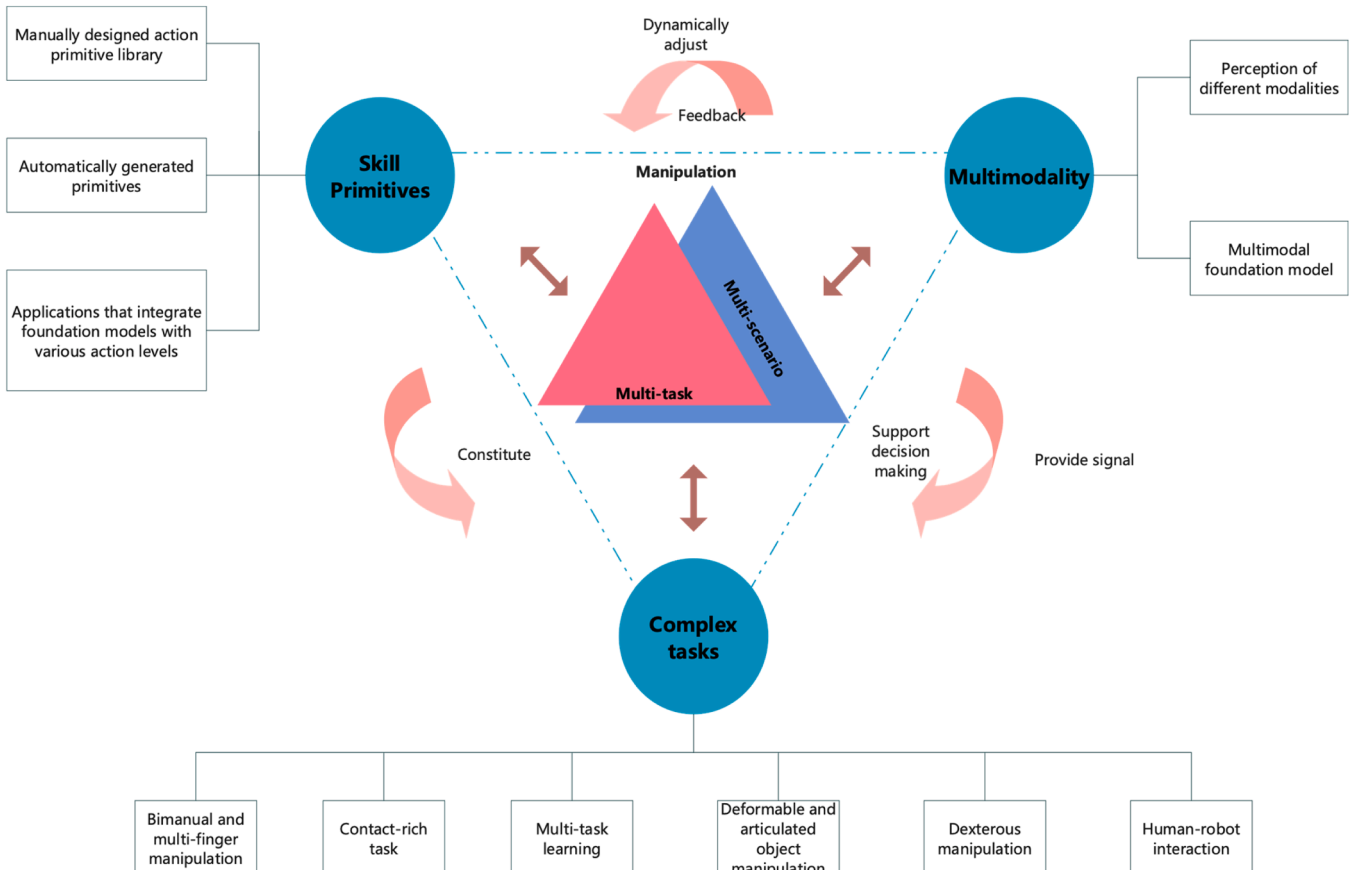


Fig. 4. Overall structure of manipulation.

perform novel tasks. Researchers are currently exploring the utilization of LLM guidance to facilitate the learning of action primitives for skill completion [158,159].

Some studies focused on developing a library of action primitives and skills. The Activity Programs knowledge base [160] collected by Puig et al. has 292 different high-level tasks in the knowledge base. The dataset covers 75 atomic operations and 308 objects, making 2709 unique steps. This work lays the foundation for future generations to explore further. Huang et al. [161] randomly sampled 88 holdout tasks from the aforementioned knowledge base for evaluation, and the remaining 204 tasks were used as a demonstration set, from which they were used as examples for prompting language models. For supervised fine-tuning baselines, these tasks were utilized in fine-tuning the pre-trained language model. RFUniverse [162] is a novel physics-based action-centric environment for robots to learn daily household tasks. RFUniverse supports 87 atomic operations and interactions between 8 primitive object types in a visually and physically plausible way. To demonstrate the usability of the simulated environment, the learned algorithm is applied to various types of tasks, namely fruit picking for manipulation, cloth folding and sponge wiping, stair chasing for locomotion, room cleaning for multi-agent collaboration, milk pouring, and hand-lifting tasks for cloning behavior from a VR interface. ManiSkill2 [163] contains a heterogeneous collection of 20 task families. A task family represents a series of task variants that have the same goal but are associated with different assets and initial states. ManiSkill2 covers different types of manipulation tasks: rigid/soft body, fixed/mobile base, and single arm/double arm.

Recent studies have focused on developing frameworks for automatically generating manipulation primitives [164]. For example, the theory and computation of contact patterns classify and enumerate manipulation primitives. The contact patterns form a graph, and the Automatically Generated Operating Primitives (AMP) algorithm performs graph-based optimization on the contact pattern mesh and solves a linear program to generate each primitive. Experiments demonstrate the robustness of the method to real-world modeling errors.

Research has also been conducted on applications that integrate foundation models with various action levels [165]. In addition, Brohan et al. [80] exploited a dataset collected by 13 robots over 17 months, containing approximately 130k episodes and over 700 tasks. The authors counted the number of verbal instructions the system could execute, where one instruction corresponded to a verb surrounded by one or more nouns, such as "Put the water bottle upright." The authors group instructions by the verbs used in them, which they call skills. Current skills include picking, placing, opening, and closing drawers, placing items in and out of drawers, holding long items upright, knocking them over, pulling napkins, and opening jars. In contrast, Vaucher et al. [36] extracted 28 kinds of atomic operations for synthetic operations of chemical experiments using three models, which are rule-based action extraction from Pistachio, rule-based action extraction from NLP, and a combination of the above two methods. Wu et al. [166] propose to use pick-and-place and pick-and-throw as action primitives, arguing that they are well suited for household cleaning, and thus propose a personalized robot assistance model with a large language model. Furthermore, there are additional pertinent works that can be referenced [145,167–169].

3.1.2.2. Learning of Complex Manipulation Tasks for Robots . Tasks are the combination of skills in time and space. Complex tasks are high-level tasks achieved by combining and scheduling multiple action primitives, representing the overall manipulation of the robot in a specific scenario. With the expanding scope of application scenarios and increasing performance requirements for robots, the demand for completing increasingly complex and precise manipulation tasks has risen. Therefore, the research goal is to empower robots with the capability to autonomously learn from their environment and independently accomplish complex

manipulation tasks. This includes acquiring knowledge, accumulating experience, and continuously updating and expanding manipulation skills in order to execute complex tasks successfully.

As an important branch in the field of robotic manipulation, dual-arm multi-finger manipulation refers to the ability of robots to perform complex manipulation using two robotic arms and multi-finger manipulators. The high degree of freedom and complexity of manipulation face challenges. LABOR [170] proposed bimanual orchestration based on language models, an agent that uses LLM to analyze task configurations and design coordinated control strategies to solve long-distance bimanual tasks. BTLA [171] introduced bimanual teleoperation based on a large language model assistant framework. The framework adopts LLM to drive the robot partner, enabling it to assist the human operator with higher accuracy and provide real-time feedback based on voice commands. Bi-VLA [172] proposed a two-handed robot dexterous operation system based on the vision-language-action model. VoxAct-B [173] is a language-conditioned voxel method that uses a vision-language model (VLM) to prioritize key areas in the scene and reconstruct a voxel grid. This voxel grid provides a bimanual manipulation strategy to learn actions and stabilize actions. This method can learn policies from voxels more effectively and can be generalized to different tasks. DexFG [174] learns human-like functional grasping of multi-fingered hands with a small number of demonstrations. HATO [175] is a low-cost arm teleoperation system that explores learning from human demonstrations using a two-handed system with multi-fingered hands and visual-tactile data. Morihira et al. use off-policy reinforcement learning and temporal contrastive learning to achieve touch-based manipulation of multi-fingered robots [176].

Rich contact refers to the complex physical interactions of a robot with an object or environment through multiple contact points when performing a task. Learning general strategies to perform various contact-rich manipulation tasks has been studied in detail in robotics, but how to find general methods for these tasks remains a difficult problem. GenChiP [177] exploits the world knowledge in LLM and combines it with the appropriate task action space. With the right action space, LLM can successfully generate strategies for various contact-rich and high-precision manipulation tasks, even under noisy conditions. Simeonov et al. proposed multimodal rearranged relational pose diffusion [178]. ChainedDiffuser [179] unifies action key pose prediction and trajectory diffusion generation to learn robot manipulation from demonstrations.

The ability of multi-task robotic manipulation to perform complex tasks and adapt to new scenarios remains challenging. Several studies are working to improve this difficulty. DNAct [180] uses diffusion to guide multi-task 3D policy learning. GNFactor [181] uses generalizable neural feature fields for multi-task real-world robotic learning. Octo [182] trains a general robotic policy on the Open X-Embodiment dataset based on a large Transformer. Diffusion policy [183] is a new approach to generating robot behaviors by representing the robot's visual motion policy as a conditional denoised diffusion process. Diffusion models have recently been used to solve a variety of different control tasks. 3D Diffuser Actor [184] is a 3D robotic manipulation policy with action diffusion. The method can learn real-world multi-task manipulation in 12 tasks from a few real-world demonstrations. RVT [185] is a robotic view transformer for 3D object manipulation. RVT can perform a variety of manipulation tasks in the real world with only a few demonstrations per task. VPDD [186] is a video-based discrete diffusion policy learning framework that demonstrates impressive generalization capabilities thanks to human video prediction. Compared with directly using ChatGPT to generate code, the RobotGPT [187] framework uses ChatGPT as an expert to train a reliable agent, which significantly improves the task success rate.

Deformable object manipulation and articulated object manipulation represent two important challenges in robotic manipulation technology. Deformable manipulation requires robots to be flexible and adaptable in dynamic environments, while articulated object manipulation requires

robots to have precise multi-degree-of-freedom control capabilities. Xiong et al. proposed a full-stack system for adaptive learning in an open-world environment to manipulate a variety of articulated objects. The system can learn from very few online samples. The system can also learn using rewards from VLM [188]. FlowBot3D [189] learns 3D joint flows to manipulate articulated objects. ManiFoundation [190] develops a foundation model for general robotic manipulation, formalizing manipulation tasks as contact synthesis, manipulating articulated rigid objects, rigid objects, and deformable objects in different dimensions. CAMS [191] proposes a two-stage framework to synthesize human-like manipulation animations. The framework achieves state-of-the-art performance in both rigid and articulated categories. The versatility of diffusion-based imitation learning (DIL) is combined with the high-quality motion generation capabilities of probabilistic dynamic motion primitives (ProDMPs), focusing on gentle manipulation of deformable objects [192]. RoboCraft [193] learns to observe, simulate, and shape elastic-plastic objects using graph networks. SAGE [194] introduces a new framework for language-guided articulated object manipulation. The authors also study the combination of general large-scale vision and language models with domain expert models to enhance the richness and correctness of network predictions, better handle this task, and achieve state-of-the-art performance.

Dexterous manipulation refers to the robot's ability to operate objects with high precision and flexibility in a small space. Robotic manipulation technology has made significant progress in the past few decades. However, complex and precise dexterous manipulation tasks remain a major challenge in the field of robotics. EUREKA [44] proposes a human-level reward design algorithm powered by LLM, demonstrating for the first time a simulated shadow hand capable of performing pen-turning skills and expertly turning a pen rapidly. Ding et al. learned universal human priors for dexterous manipulation from human preferences [195]. Extensive research has also been conducted on dexterous manipulation [196–214].

Human-robot interaction refers to the process of interaction or collaboration between robots and humans in physical manipulation tasks. Although some explorations have been made by predecessors, there are still challenges in natural interaction capabilities, safety, adaptive learning, etc. Some studies have built frameworks for interactive, natural language-guided robots in the real world [215,216]. Liu et al. enhanced the manipulation capabilities of LLM-based robots through human-robot collaboration [217]. LaMI [218] seamlessly integrates LLM with robotic systems to enhance human-robot interaction. After implementation on a physical robot, LaMI demonstrated the ability to adapt to multimodal inputs and dynamically interact with humans through voice, facial expressions, and GUI modules. SeeAsk [219] proposes an interactive disambiguation system for open-world robotic grasping tasks. With the help of a large-scale pre-trained visual language model, SeeAsk can plan to ask a variety of questions (self-referential, relational, or pointing-based) for effective interaction or decide to grasp. Lynch et al. proposed a robot framework for real-time natural language interaction in a real environment. The relevant resources are open source (datasets, environments, benchmarks, and policies). Behavioral cloning training is performed on the dataset of annotated trajectories, and the generated policies proficiently execute orders of magnitude more commands than previous work [220].

In addition, some outstanding work deserves attention, such as manipulation trajectory planning (LATTE [221], VoxPoser [222]), skill acquisition (RoboGen [223], DIAL [145]), zero-shot generalization manipulation (Track2act [224], SuSIE [225], RoboFlamingo [226], PROGRAMPORT [227]), an end-to-end framework for fine-grained manipulation (CLIPORT [228]), and end-to-end video conditional policy learning (Vid2Robot [229]).

3.1.2.3. Robot Manipulation Learning with Multimodal Fusion. Multimodal perception and interaction provide the necessary information and

feedback support, enabling robots to make accurate decisions and dynamic adjustments when performing complex tasks, ensuring the smooth completion of tasks. While multimodality [230] addresses the challenge of agent perception, it falls short in addressing the issue of cognition. In various scenarios, such as intelligent customer service, users often provide information through multiple modalities. Multimodality holds significant perceived value, but it still has substantial challenges to overcome in solving fundamental issues. Multimodality represents a prominent future trend wherein foundation models will progressively align with multimodal approaches, enabling future agents to operate within a multimodal environment [34,231,232].

Many studies have been conducted between different modalities or combinations of different modalities, such as force, touch, smell, audio, image, text. Yang et al. proposed MOMA-Force, an imitation method for visual binding force, using perceptual representation learning, imitation learning, and admittance-based whole-body control to enhance the robustness and controllability of the system. MOMA-Force enables mobile manipulators to learn multiple complex contact-rich tasks with a high success rate and small contact force [148]. Jiang et al. demonstrate that a wide range of robotic manipulation tasks can be expressed with multimodal prompts, and many robotic manipulation tasks can be represented as multimodal prompts interleaved with language and images or video frames. The authors design a transformer-based robotic agent, VIMA, that processes these prompts and outputs motor actions autoregressively [233]. Sferrazza et al. achieve robot generalization manipulation through multimodal learning that fuses vision and touch. AudioPaLM combines text-based and speech-based language models [134]. Brohan et al. propose to jointly fine-tune state-of-the-art vision-language models on robot trajectory data and Internet-scale visual-language tasks. The authors present RT-2, a vision-language-action model that transfers network knowledge to robot control [141]. RobotGPT provides the perception, cognition, decision, and execution capabilities with embodied AI [234]. Luo et al. proposed a physics-based humanoid controller that enables high-fidelity motion imitation and fault-tolerant behavior in the presence of noisy inputs (e.g., pose estimates generated from videos or speech) and unexpected falls [235]. Gandhi et al. conducted the first large-scale study of the interaction between sound and robot movement. The authors created the largest available sound-action-vision dataset using the robotics platform Tilt-Bot, with 15000 interactions on 60 objects [99]. Peng introduces Kosmos-2, a multimodal large language model (MLLM) that supports perceptual object descriptions (such as bounding boxes) and new capabilities for integrating text into the visual world [236]. Radosavovic et al. proposed an RPT model, which is a Transformer that operates on a sequence of sensorimotor tokens. Given a sequence of camera images, proprioceptive robot states, and past actions, it is assumed that if the robot can predict what is missing, then it has acquired a good model of the physical world that allows it to act [237]. Li et al. systematically studied how visual, auditory, and tactile perception can jointly help robots solve complex operating tasks [238]. LEE et al. use graph neural networks to generate master odor maps (POMs) that preserve perceptual relationships and enable odor quality predictions for previously uncharacterized odors [239]. Guzey et al. proposed visually stimulated tactile adaptation in order to learn tactile flexibility, thereby correcting errors and adapting to changing situations [240]. Hearing Touch [241] is audio-visual pre-training for rich contact manipulation. Contact microphones replace tactile sensors to capture inherent audio-based information. Large-scale audio-visual pretraining obtains representations that improve robot manipulation performance. HDP [242] is a hierarchical agent for kinematically perceptive robotic manipulation. Lin et al. [175] explore learning from human demonstrations using a two-handed system with multi-fingered hands and visual-tactile data.

In addition, there are studies exploring large multimodal models for robotic manipulation. Li et al. proposed ManipLLM [243], an embodied multimodal large language model for object-centric robotic manipulation. Huang et al. proposed ManipVQA [244] to inject robot capabilities

and physical grounding information into multimodal large language models. Octopi [463] is a system that leverages tactile representation learning and large vision-language models to predict and reason about tactile input, requiring minimal language fine-tuning. GLaMM [464] is the first model capable of generating natural language responses interleaved with object segmentation masks, enabling enhanced multimodal user interactions.

3.1.3. Datasets and metrics

Datasets. In order to improve the generalization ability of robot manipulation learning, researchers collected multi-task, multi-scenario datasets so that the robot can be trained to quickly generalize various scenarios. At the same time, the test datasets evaluate the manipulation capabilities of these robots guided by the foundation model. There are several datasets available for testing, as presented in Table 4.

Metrics. In order to improve robot performance and manipulation level, the key is to continuously optimize the performance of the robot learning system. Then, the core issue is to reasonably evaluate the performance of the robot. Issues encountered in the comprehensive evaluation process: improvement of single evaluation indicators, integration of multiple evaluation indicators, weight distribution of each indicator, and reliability of evaluation results.

Manipulation performance is evaluated by (1) Plan success rate, which measures whether the skills selected by the model are correct for the instructions, regardless of whether they are actually successfully executed. (2) Execution success rate, which measures whether the entire system actually successfully executed the required instructions [80, 141].

3.1.4. Problems

Despite making considerable progress, significant problems still exist. Robot movements are continuous and complex, with more physical interactions and manipulation causality.

A comprehensive library of skills and actions needs to be built. In order to complete downstream target tasks, a skill library needs to be built. High-level instructions are broken down into atomic actions. The existing skill action library suffers from a long-tail distribution problem. A more comprehensive skill action library needs to be established to assist robot skill learning. Robots can manipulate objects of any size, shape, and material in any scene. For example, a robot can pour a specified volume of any liquid and complete specified tasks in a dynamically changing environment. Objects that are difficult to manipulate in the 3D world, such as articulated, deformable, and transparent objects, can be manipulated. The robot can solve the problems of occlusion and obstacle avoidance when grasping. The robot can reason about spatial relationships and contact dynamics, accurately drive high-degree-of-freedom arms, and apply the appropriate force to

stably grasp objects without breaking them.

Current robotic object manipulation applications rely heavily on human programming and have poor autonomy, which is unrealistic in the long run. Vision-based intuitive physics, open-set grasping with semantic representation, and planning under partial observability and uncertainty will be the future trends of robot grasping [17].

3.2. Navigation

3.2.1. Characteristics and challenges

Navigation involves the robot's positioning and path selection in the physical environment. It includes obtaining environmental information based on sensors (such as lidar, cameras) and calculating the optimal route from the starting point to the target point. The core function is to provide position positioning and obstacle avoidance functions. Several research projects focus on developing autonomous robots capable of navigating in different environments. However, little research has specifically addressed the navigation challenges faced by legged robots in indoor environments, especially when relying on single-camera vision. The legged robot is able to traverse uneven surfaces and overcome obstacles, such as stairs, that traditional wheeled robots typically cannot reach. In order to realize the autonomous navigation of mobile robots in unknown dynamic environments, mobile robots need to solve three basic problems, namely "Where am I?" "What are my surroundings?" and "What should I do next?". There are three core challenges to achieving autonomous robot navigation, which can be divided into map-based navigation and map-free navigation. In order to solve these three core basic problems, a series of core technologies such as environmental perception, autonomous positioning, and motion planning are required [257,258]. Foundation models have reasoning and decision-making capabilities, and multi-modal foundation models can handle rich information modalities and encode information of different modalities into the same vector space, which is conducive to cross-modal information processing. Benefiting from these considerations, foundation models help to deal with problems such as motion planning and environment perception of robots. This article only discusses navigation with maps. For navigation without maps, refer to the references [259]. In this section, we survey recent research results and state-of-the-art advances in the direction of robotic navigation (Fig. 5) with foundation models, including robotic environment perception, map creation, autonomous localization, and motion planning [260].

In general, the logical relationship between these four steps is as follows. First, environment perception is to collect surrounding information and provide it to map creation, autonomous localization, and motion planning. Secondly, an environmental map is generated based on environment perception data for localization and motion planning. Then, the current position of the robot is calculated based on the

Table 4
Manipulation Datasets.

	Objects	Scene	RGB-D Images	Depth Images	3D point clouds	Size	Num of grasp	6 DoF	Multi-cam	Real Image	Humanoid gripper
DailyGrasp [245]	26	-	26005	26005	-	26k motion steps	-	7DoF	✓	✓	×
Stanford Grasping [246]	10	-	13747	13747	-	-	-	✓	-	×	-
Cornell Grasping [247]	240	-	885	885	-	7200 grasps	-	✓	-	✓	×
YCB Benchmarks [248]	77	-	46200	46200	✓	-	-	✓	✓	✓	×
CMU dataset [249]	150+	-	50567	-	-	About 1 G	50 k	7DoF	✓	✓	×
Google grasping dataset [89]	-	-	2898410	-	-	-	-	7DoF	✓	✓	×
Dex-Net 4.0 [250]	1664	-	5 M	5 M	✓	103.16 G	-	-	×	×	×
JACQUARD [251]	11k	-	54485	108970	-	-	1.1 M	-	×	×	×
GraspNet-1Billion [252]	88	190+	97280	-	-	2.1B	1B+	✓	✓	✓	×
kPAM-SC [253]	33	125	111k+	-	✓	111k	-	7DoF	×	✓	×
Min Liu et al. [254]	324	-	-	-	-	-	-	25DoF	×	×	✓
HO14D [255]	800	610	2.4 M	2.4 M	✓	-	-	-	✓	✓	✓
DexGraspNet [201]	5355	-	-	-	-	-	1.32 M	-	-	×	✓
Grasping-anything [256]	~ 3 M	-	-	1 M	-	1 M samples	~ 600 M	-	-	×	-

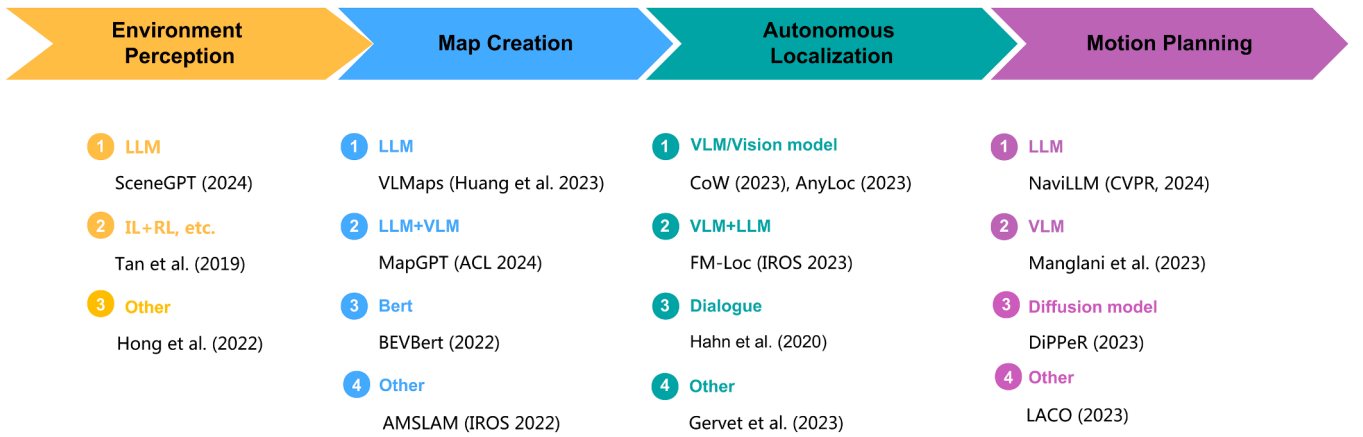


Fig. 5. Overall structure of navigation.

matching of perception data with the map. Finally, motion planning uses the autonomous localization information of the map to plan the optimal path and adjust it in real time.

3.2.2. Methods

3.2.2.1. Environment Perception. Environmental perception refers to the process in which a robot, during its movement, perceives the surrounding environment through various devices. It can collect real-time environmental information and adjust its position and pose accordingly based on the actual situation.

Some studies have investigated this challenge using foundation models. SceneGPT [261] develops a 3D scene understanding system based on LLM without the need for 3D supervised data and large-scale training mechanisms. The model demonstrates that simple techniques such as contextual cues can help tune LLM and tap into its spatial reasoning capabilities. SCENEVERSE [262] extends 3D visual language learning for solid scene understanding. SQA3D [263] is a benchmark that combines the advantages of contextual understanding and contextual reasoning to study the ability of embodied scene understanding. Methods that align the embedding space of different modalities (in this case, IMU data) with the visual embedding space enable the VLM to understand and reason about these additional modalities without the need for retraining. The results show that using multiple modalities as input can improve VLM's scene understanding and enhance its overall performance in various tasks [264].

Some research is based on the extension of imitation learning and reinforcement learning. Tan et al. use environment dropout for back-translation to learn unseen environments in navigation [265]. Qi et al. propose an object and action perception model that can flexibly match object-centered or action-centered instructions with their corresponding visual perception or action direction in the process [266]. Various other studies can serve as references [267–269].

There are also other ways to explore learning environment information. The agent follows real-life navigation instructions, then recognizes the location described in natural language and finds the object at the target location [270]. Hong et al. proposed a predictor to generate a set of candidate waypoints during navigation, bridging the gap between learning in discrete and continuous environments for visual and language navigation [271].

3.2.2.2. Map Creation. Map construction can be accomplished either through the utilization of metric maps or by employing symbols that represent the robot's position in its reference frame.

Some studies explore map construction based on foundation models. To address the lack of spatial accuracy of classical geometric maps, Huang et al. proposed spatial map representation VLMs, which fuses

pre-trained visual-linguistic features with 3D reconstructions of the physical world. VLMs can be autonomously constructed from the robot's video sources using standard exploration methods and can support natural language indexing of maps without labeled data [272]. An et al. proposed a multi-modal map pre-training paradigm BEVBert for language-guided navigation, which has spatial awareness capabilities [273]. MapGPT [274] constructs maps online using map-guided prompts. These prompts provide GPT with node information and topological relationships to activate global exploration. In addition, the authors propose an adaptive planning mechanism that performs multi-step path planning based on the map, enabling the agent to systematically explore potential goals.

There are also other ways to explore this challenge. Jia et al. proposed a neural SLAM method that, for the first time, exploits multiple modes to explore, predict perceptible semantic maps, and plan them simultaneously [275]. Zhang et al. propose a hierarchical object-to-region (HOZ) environment map to guide the agent in a coarse-to-fine manner. Additionally, an online learning method is proposed to update the HOZ new environment based on real-time observations [276].

3.2.2.3. Autonomous Localization. Robot localization refers to a robot's ability to accurately determine its position and orientation within a predefined reference frame.

Recently, there has been research on using large language models combined with vision-language models for localization. FM-Loc [277] is a new image-based localization method based on foundation models. It uses a large language model, GPT-3, combined with a vision-language model, CLIP, to build a semantic image descriptor. The descriptor is robust to severe changes in scene geometry and camera viewpoint.

Some studies use vision-language models such as CLIP to address this challenge. Inspired by the recent success of open-vocabulary models for image classification, Gadre et al. investigate a simple framework, CLIP on Wheels (CoW), to adapt open-vocabulary models to language-driven zero-shot object navigation without fine-tuning (L-ZSON) [278]. Keetha et al. have developed a universal VPR solution - a technology that works across a variety of structured and unstructured environments without the need for any retraining or fine-tuning [279].

Some studies obtain localization information through dialogue. Based on the WAY dialogue dataset, Hahn et al. focus on the LED task (locating the observer from the dialogue history) [280]. Two agents (tourist and tour guide) interact through natural language to allow tourists to navigate to the correct location [281].

There are also some other methods for localization that are worth referring to. Khandelwal et al. built a simple baseline EmbCLIP without task-specific architectures, inductive biases (e.g., using semantic maps), auxiliary tasks during training, or depth maps, but the improved

baseline performs remarkably well across a range of tasks and simulators [282]. Gervet et al. proposed a semantic visual navigation method, compared six typical approaches of classical, modular, and end-to-end learning methods without any experience, maps, or instruments, through large-scale empirical research, found that modular learning is a reliable way to navigate objects [283].

3.2.2.4. Motion Planning. Motion planning is the process by which a robot intelligently selects an optimal sequence of actions to move toward a target location based on its current position within a reference frame, considering changes in the environment.

There is a lot of work that combines large language models with navigation planning. Zheng et al. proposed the first general model for embodied navigation, NavILLM [284], which adapts LLM to embodied navigation by introducing pattern-based instructions. The approach enables the integration of different data sources from various datasets into the training, giving NavILLM a wide range of capabilities required for embodied navigation. LM-Nav [142] is a system for robot navigation via textual instructions that can control a mobile robot without any user annotation of the navigation data. The approach enables mobile robots to navigate long distances in complex outdoor environments based on natural language instructions. NavCoT [285] implements parameter-efficient in-domain training, enabling LLM to perform self-guided navigation decisions. NavGPT [286] is a purely LLM-based instruction-following navigation agent. Experiments show that NavGPT can explicitly perform high-level planning for navigation and perform zero-shot sequential action prediction for visual and language navigation. SayNav [287] gradually builds a 3D scene graph of the explored environment and converts it into an LLM to generate dynamic and contextual high-level navigation plans. This is a new method that efficiently generalizes to complex navigation tasks in unknown large-scale environments. Qian et al. proposed the March-In-Chat model, which allows real-time communication with LLMs and performs dynamic planning based on room-and-object aware scene perceiver. The model outperforms previous models on the REVERIE benchmark [288].

There is work that applies vision or visual language methods to path planning for navigation. Mangani studies the development of visual segmentation and path-planning algorithms for autonomous navigation and obstacle avoidance systems in domestic environments [257]. Lin et al. propose to provide action prompts for VLN agents to enable explicit learning of action-level modality alignment for navigation. Specifically, action prompts are defined as a pair of modality-aligned image sub-prompts and text sub-prompts, and when navigation starts, a set of action prompts related to an instruction is retrieved from a pre-built library of action prompts and passed through a prompt encoder to get hint features. Then, the prompt features are concatenated with raw instruction features and fed to a multi-layer transformer for action prediction. Using a contrastive language-image pre-training (CLIP) model, a modality alignment loss and a sequential consistency loss are introduced to enhance the alignment of action prompts and force the agent to sequentially concentrate on related prompts [289].

Some work applies diffusion models to motion planning for robots. DiPPeR [290] is a novel 2D path planning framework for fast quadrupedal motion that leverages diffusion-driven techniques. The planner successfully generates feasible paths of variable length 80 % of the time on average for maps of various sizes and obstacle structures. Successfully tested on two different robots, the planner validates its transferability to the real world and demonstrates platform-independent capabilities. Chang et al. [291] introduced an end-to-end approach that employs a conditional diffusion model to generate achievable goals that avoid collisions using only visual input. Carvalho et al. [292] proposed diffusion of motion planning, which uses diffusion models to learn and plan robot motion.

There are also some works on large-scale video-based visual

language models for navigation. In order to enable agents to navigate in unseen environments following language instructions, NaVid [293] proposed a large-scale video-based visual language model to alleviate this generalization gap. NaVid only needs a live video stream from the robot's monocular RGB camera to output the next action. NaVid is the first attempt to demonstrate the ability of VLM to achieve state-of-the-art navigation performance without any map, odometer, or depth input.

There is work combining language with reinforcement learning for motion control. LAGOON [294] is language guided motion control. It is a multi-stage reinforcement learning approach that generates physically realistic robot actions given language commands.

There is some particularly novel work incorporating contact awareness into language-conditioned path planning. Traditional path-planning algorithms focus only on collision-free paths, limiting their applicability to contact-rich tasks. To address this challenge, LACO [295] proposes the field of language-conditioned path planning, where contact awareness is incorporated into the path planning problem. This is a new approach to learning collision functions using only single-view images, language prompts, and robot configurations. Experiments show that this approach can facilitate complex, nuanced path planning that allows interactions with collision-safe objects rather than prohibiting any collisions.

There are other motion planning methods. N²M² [296] extends the kinematic feasibility formulation of mobile operations to complex unstructured environments. This work focuses on achieving arbitrary motion and uses a simple robot-independent end-effector motion generator to perform unseen long-horizon tasks in unexplored environments while immediately reacting to dynamic obstacle avoidance and environmental changes.

3.2.3. Datasets and metrics

Datasets. In order to promote the development of robot navigation tasks, researchers have collected a large amount of data from multiple aspects such as instructions, scenes, objects. This plays a very important role in the development of navigation and also lays the foundation for the transfer from virtual to real environments. There are many datasets that can be used for navigation. See Table 5 below.

Metrics. Navigation performance is evaluated by (1) Success / Oracle Success Rate (%). (2) Navigation Error (m). (3) SPL (Success weighted by Path Length). (4) CLS (Coverage weighted by Length Score), measuring fidelity to the reference path. (5) nDTW (normalized Dynamic Time Warping). (6) SDTW (Success weighted by normalized Dynamic Time Warping) [299,303,305].

3.2.4. Problems

To solve the problem of high obstacles, it is worth considering investigating partially observable search methods. Potential solutions would involve broadening the scope of the study to cover a wider range of domestic environments (including stairs and uneven surfaces), outdoor environments, and complex unknown environments. In addition, a more general large-scale navigation model can be designed that can be widely applied to any robot, similar to LLMs and VLMs for processing any text or image.

In addition, the efficiency of multi-target navigation needs to be improved, scene knowledge needs to be better applied in navigation, and the efficiency of unknown scene navigation needs to be improved. Techniques for optimizing system efficiency in real-time scenarios will also be explored, taking into account factors such as robot stability, dynamic obstacle avoidance, and resource constraints.

Path planning in unknown scenarios faces the following problems: lack of a global map, generalization of unknown environments, and sensor noise in real-life applications. The lack of a global map includes the following aspects: lack of a global SLAM map, how to learn prior knowledge of layout, and in what form to store knowledge. The generalization of unknown environments includes the following aspects:

Table 5
Visual language navigation datasets.

	Data type	Size	Scenes	Annotation	Objects	Instructions	Buildings
NavigationNet [297]	Image	Hundreds of thousands of images	15	-	-	-	Each scene with 1–3 rooms
M2DGR [87]	Image, IMU, VI sensor, LiDAR, GNSS	1220.6 GB	36	-	-	-	-
DISCOMAN [298]	Image, depth, IMU, grid	200 long sequences	-	-	-	-	-
ALFRED [299]	Images, instructions	8055 expert demonstration episodes	120	25k	-	25k	-
VXN [300]	Image, text, audio	9.6 M episodes	58	-	21 (categories)	-	-
SCAND [301]	Rosbags, LiDAR, Images, Visual Odometry, Wheel odometry	8.7 h/138 trajectories	-	1656	-	-	-
MuSoHu [302]	Image, LiDAR, IMU, audio, depth	300 trajectories	-	17 labels	-	-	-
R2R [303]	Image, text	7189 trajectories	90	-	-	22k	-
BnB [304]	Image, text	1.4 M	140k	0.7 M	-	-	-
REVERIE [305]	Image, text	10318 panoramas	132	-	4140	21702	86

limited training rooms and infinite unknown environments, as well as the transfer and generalization of knowledge. Sensor noise in real-life applications includes the following aspects: depth loss, RGB blur, and detection and segmentation errors.

3.3. Task planning

3.3.1. Characteristics and challenges

Task planning is the design of a series of manipulation steps or actions to complete the target task. The core function is to calculate and decide the order and method of actions based on the target, environment and constraints. Depending on the difficulty of the task, the task can be completed by a single robot or by multiple robots. Many of these problems require multiple robots to work together in multi-robot systems. Long-horizon tasks for a single robot need to be broken down into smaller tasks that can be accomplished with simple operations. Task planning for multiple robots working together needs to consider the relationship between tasks, robot capabilities and cooperation, and other challenges. In addition, the agent cannot hallucinate when planning tasks. Within the bounds of common sense, agents can learn from previous tasks, self-correct and improve, and continue to learn. This is a challenging task. The true realization of intelligent robots will be based on the perception of the environment, dynamic learning, and continuous updating. In this section, we survey the recent research results and the latest progress of foundation models in the direction of robot task planning (Fig. 6), including single-robot task decomposition, multi-robot task planning, self-correction, self-improvement, and continuous

learning of agents.

Structurally, a single robot has developed into multiple robots; evolutionarily, robots that can complete tasks have evolved into robots that can self-correct and improve.

3.3.2. Methods

3.3.2.1. Single-Robot Task Decomposition. The best solution for complex tasks is to split them into several simple tasks and then solve them one by one. Generally, the effect will be good. Foundation models have shown good zero-shot generalization ability. State-of-the-art chatbots can provide plausible answers to many common questions that arise in everyday life. However, so far, foundation models cannot reliably solve long-horizon planning problems [306].

In order to solve the long-horizon planning problem, many research works are based on the planning framework of the language model. Liu et al. introduce LLM+P, the first framework to incorporate the strengths of the classical planner into LLM [307]. Lin et al. proposed a language-based planning framework, Text2Motion, based on which robots can solve sequential manipulation tasks for long-horizon reasoning. This framework uses the feasibility heuristic encoded in the skill library Q function to guide the task planning of LLMs, and Geometric dependencies among skills are resolved by performing geometric feasibility planning during the search process [308]. Wake et al. use ChatGPT to convert natural language instructions into executable robotic actions in long-step scenarios [309]. TPTU proposed a structured framework customized for LLM-based artificial intelligence agents, within which

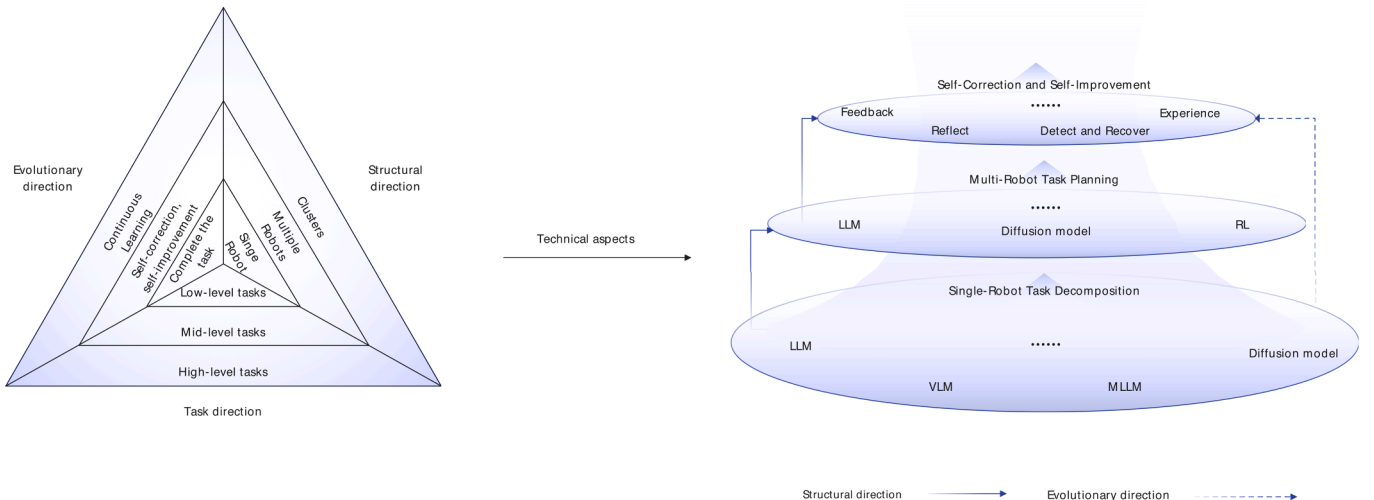


Fig. 6. Overall structure of planning.

two different types of agents were designed, one-step agents and sequential agents, to perform the reasoning process. The framework is then instantiated using LLMs and evaluated on their task planning and tool usage abilities on typical tasks [310]. Zhen et al. propose a task-planning method that combines human expertise with LLMs. The LLM prompt template is designed to have a stronger expressive ability to represent structured professional knowledge. Further, a method of gradually decomposing tasks to generate task trees is proposed. A strategy for decoupling robot task planning is designed [311].

When solving planning problems, language models can provide rich semantic knowledge and generate complex plans, but they lack real-world information and usually produce infeasible action sequences. Therefore, some work considers incorporating physical scene constraints and performing task planning based on vision-language models. Wu et al. aligned the LLM with the visual perception model and generated executable action sequences based on the objects existing in the real scene [312]. Obinata et al. describe a strategy for implementing a robotic system capable of performing general service robot (GPSR) tasks in robocup@home [313]. The authors propose a general service robot system that integrates a large language model, visual language models, and state machine task executables. The state machine task executable can clearly define the robot's actions when it succeeds or fails, making its actions deterministic and stable in real environments.

Embodied AI is an important frontier in robotics technology, which can plan and execute robot action sequences to complete long-range tasks in a physical environment. Some research works have proposed an end-to-end multimodal foundation model based on embodied AI to provide multimodal understanding and execution capabilities for embodied robots. EmbodiedGPT introduces an efficient training method to generate high-quality plans. The paradigm of extracting task-relevant features from LLM-generated planning queries is introduced to form a closed loop between high-level planning and low-level control. The authors point out that extensive experiments show that EmbodiedGPT is effective on embodied tasks, including embodied planning, embodied control, visual captioning, and visual question answering [314].

A large category of methods also apply foundation models, such as diffusion models, to robotics for planning. Diffusion models have emerged as a promising data-driven planning method and have demonstrated impressive capabilities in robotic control, reinforcement learning, and video planning. Video language planning [315] is one method available. SkillDiffuser [316] is based on interpretable hierarchical planning of skill abstraction in diffusion task execution. SafeDiffuser [317] uses a diffusion probabilistic model for safety planning. There are also some works on diffusion-based embodied task planning frameworks under uncertainty [318]. Leveraging diffusion models for long-horizon skill planning, the authors introduce generative skill chains [319], a probabilistic framework that learns skill-centric diffusion models and combines their learning distributions to generate long-horizon plans during inference. EDGI [320] uses embodied agents for equivariant diffusion of planning. Compositional foundation model [321] of hierarchical planning (HiP): The lowest level of planning performs visual reasoning and executes actions according to the designed plan through visual motion control. Multiple expert foundation models are used to separately train language, visual, and action data to jointly solve the long-horizon task. Cold diffusion on the replay buffer learns to plan from a known good state [322]. Using adaptive online replanning of diffusion models, an important question addressed is when a given plan should be regenerated due to action execution errors and changes in the external environment [323].

Different from the above methods, there are other methods to solve the long-term planning task. Gu et al. studied a modular approach to handle the long-view movement operation task for object rearrangement, decomposing the complete task into a series of subtasks [324]. RoboCook developed an intelligent robotic system that can use multiple tools for long-horizon elastoplastic object manipulation. RoboCook uses point cloud scene representation, uses graph neural networks (GNNs) to

model tool-object interactions, and combines tool classification with self-supervised policy learning to develop action plans. It demonstrates robustness to severe external disturbances and exhibits adaptability to different materials [325].

3.3.2.2. Multi-Robot Task Planning. For multiple robots, task planning needs to consider the relationships between tasks, robot capabilities and cooperation, and other challenges. Task planning for multi-robot teams usually involves three sub-problems: task decomposition, task allocation and task scheduling, which are introduced here to consider situations where robots sequence their own tasks to satisfy various constraints. While each of these problems can be solved individually, the interdependencies between these subproblems are often considered in the context of integrated tasks to provide solutions that are often better than solving each subproblem independently [20,326,327].

Foundation models have shown impressive planning capabilities in single-agent-specific tasks across domains, yet their planning and communication capabilities in multi-agent cooperation remain unclear. A novel framework for multi-agent cooperation using LLMs is proposed, which enables agents to plan, communicate, and cooperate with other agents or humans to effectively complete long-horizon tasks. Experiments have proven that LLM-based agents that communicate by language can win more trust and cooperate more effectively with humans [328]. RoCo is a unified approach for multi-robot collaboration that leverages pre-trained LLMs for high-level communication and low-level path planning. The authors introduce the RoCoBench benchmark, which includes various challenges in collaborative task scenarios. The method demonstrates its utility on RoCoBench [329]. ChatDev divided the development process into four different chronological stages: design, coding, testing, and recording. Each stage involves a team of agents, such as programmers, code reviewers, and test engineers, facilitating collaborative conversations and a seamless workflow. The chat chain acts as a facilitator, breaking down each stage into atomic subtasks. This paradigm demonstrates how linguistic communication facilitates multi-agent collaboration, establishing language as a unifying bridge for autonomous task-solving between LLM agents [330].

Using multi-agent competition, the simple goal of hide-and-seek, and large-scale standard reinforcement learning algorithms, Bowen Baker et al. found that the agent created a self-supervised autocurriculum, triggering multiple rounds of different emergent strategies, many of which required complex tool use and coordination [331]. The authors discuss how agents trained through multi-agent competition in a simple game of hide-and-seek can develop complex emergent behaviors without having to explicitly program those behaviors. The results suggest that multi-agent environments may be more effective than traditional RL in creating agents with skills applicable to real-world scenarios. Future research may explore more open environments and the possibility of transferring these learned behaviors to other domains.

Diffusion-based multi-agent research has also made some progress. MADIFF [332] is offline multi-agent learning with a diffusion model. In multi-agent scenarios, effective coordination is essential for agents to successfully accomplish teamwork by autonomously formulating the trajectory of each individual agent. MotionDiffuser [333] is a controllable multi-agent motion prediction using diffusion. This is a novel multi-agent motion prediction framework based on diffusion models capable of learning diverse, multi-modal joint future distributions of multiple agents.

3.3.2.3. Self-Correction and Self-Improvement. The ability to automatically detect and analyze failed executions is critical for explainable and robust robotic systems. This requires the agent to self-criticize, learn from previous tasks continuously, and correct behavior within the bounds of common sense. Based on the large language model, directional error correction and directional improvement can be done. In task planning, it is necessary not only to have the current state but also to

have memory, experience, reflection and summary, and world knowledge. Agents can get feedback based on actions. For large language models, reinforcement learning from human feedback is an extremely simple environment.

Many studies have explored ways to improve performance through corrections through feedback [334]. Sharma et al. show how language can be used to update the latent cost of the planner to improve task performance. This method can use the language to correct the plan in two ways: adding constraints or specifying intermediate subgoals for the planner [335]. Shinn et al. proposed a novel framework, reflexion, to learn from trial and error through dynamic memory and self-reflection to make better decisions in subsequent experiments [336]. Olausson et al. use feedback from more powerful models and human feedback to improve code self-repair capabilities [337]. CoPAL [338] uses a large language model to perform corrective planning of robot actions. The core re-planning strategy uses closed-loop multi-level feedback from the physical robot to improve executable capabilities. OLAF [339] uses verbal correction to interactively learn robot policies. Similarly, Shi et al. propose a framework that enables robots to respond to verbal corrections and improve [340]. In addition, SCMLLM [341] is a self-correcting multimodal large language model for end-to-end robot manipulation. The model has the ability to predict the posture of the end effector through fine-tuning. When the execution fails, the model can autonomously identify the error, adaptively seek feedback from experts, and correct the failed action.

The ability to leverage heterogeneous robotic experience from different robots and tasks to quickly acquire new skills has the potential to transform the way robots learn. Bousmalis et al. propose RoboCat, a generalist agent for robotic manipulation, as a visual goal-conditional decision transformer. RoboCat demonstrates the ability to generalize to new tasks and robots. It also demonstrates the use of the trained model to generate data for subsequent training iterations, providing the basic building blocks for autonomous improvement loops [147].

When faced with more complex problems that require nonlinear thinking, even the most powerful LLMs can make mistakes. To combat this, Ning et al. explore whether LLMs have the ability to identify their own mistakes without resorting to external resources. In particular, the research focuses on whether they can be used to identify individual errors in step-by-step reasoning. The authors propose a zero-shot verification scheme to identify such errors. This verification scheme is then used to perform weighted voting on different generated answers, improving question-answering performance [342].

In order to enable LLMs to integrate into the environment autonomously, Peng et al. proposed the self-driven grounding framework to automatically and gradually lay the foundation for LLMs through self-driven skill learning [343].

Explaining robot failures and generating corrective plans are critical to robotic systems. Liu et al. introduced a framework, REFLECT, to leverage the power of LLMs to explain robot failures. According to the explanation, the mission planner will generate an executable plan for the robot to correct the fault and complete the mission. Evaluating the framework on the RoBoFail dataset of failure scenarios, experiments demonstrate that the LLM-based framework is capable of generating informative failure explanations to aid in successful corrective planning [344].

The low-level execution in the physical world may deviate from the high-level textual plan due to environmental perturbations or imperfect controller design. Guo et al. proposed DoReMi, a novel language model base framework that can instantly detect and recover from inconsistencies between planning and execution [345].

The diffusion model has strong generative capabilities and has great potential as an offline reinforcement learning paradigm. However, the diffusion model is limited by the lack of diversity in training data and has defects in planning execution and task generalization. Therefore, some research improves the diffusion model and makes it a better planner. AdaptDiffuser [346] proposes the diffusion model as an

adaptive self-evolution planner.

There are some other works that can be referenced [347–354], such as WebShop [355], InterCode [356], Collie [357]. In addition, continual learning serves as a means to consistently foster self-improvement and self-optimization [358].

3.3.3. Datasets and metrics

Datasets. Researchers in the field of robot task planning rely on datasets to develop, validate, and test their planning algorithms and systems. These datasets may include real-world data collected from robots or simulated data generated within robot environments [359]. Here is an example list in Table 6.

Metrics. Task planning performance is evaluated by (1) Task Success Rate. (2) Task Durations. (3) Task Diversity. (4) Task Difficulty. (5) Task Dependencies.

3.3.4. Problems

More effective strategies for self-improvement and enhancement need to be explored [360]. Foundation models automatically detect when and how to apply the planner. Moreover, foundation models reduce reliance on human information during planning [307]. In addition, the planning time efficiency of the model needs to be improved. The correctness and executability of generated plans need to be significantly improved. Multimodal models are explored for mission planning, which can naturally support extending the planning system to higher-dimensional observation spaces.

A significant advantage of utilizing the latest foundation models is their ability to adapt to various operating environments through several learnings and user feedback. These capabilities not only eliminate the need for extensive data collection or model retraining but also allow users to make adjustments that promote safe and robust mission planning. The ability to effectively adapt to user feedback may be due in part to learning methods that combine model behavior with human intent. Additionally, the output of large models can be adjusted with a reasonable amount of feedback. The ability of foundation models to reflect the semantic content of user feedback provides a means for users to communicate their intentions to the system. Therefore, this aspect helps lay the foundation for a user-friendly system. Delving into this ability to adjust contributes to the user-friendliness of the system [309].

3.4. Reasoning

3.4.1. Characteristics and challenges

Reasoning involves understanding the complexity of tasks, changes in the environment, and making decisions under uncertainty or incomplete information. Reasoning can make reasonable judgments based on known knowledge, rules, or learned patterns. The core function is to make decisions and solve problems based on existing knowledge and data. Reasoning requires the reasoner (robot) to have an explicit representation of various parts or aspects of its environment in order to reason. Robots are increasingly transitioning from specialized single-task machines to general-purpose systems operating in diverse and dynamic environments. To solve the challenges associated with real-world interactions, robots must effectively generalize knowledge, learn, and remain transparent in their decision-making processes. However, there are few studies that specifically address the reasoning challenges faced by robots in interactive environments. This survey aims to investigate reasoning robot system technologies that enable robots to encode and use knowledge, including concepts, facts, ideas, and beliefs about the world. Continuously sensing, understanding, and generalizing knowledge enables robots to identify meaningful patterns shared across problems and environments to perform a variety of real-world tasks more efficiently [21]. Foundation models provide promising tools for robots to perform complex reasoning tasks [376]. In this section, we survey the recent research results and the latest progress of foundation models in the direction of robot reasoning (Fig. 7), including robot

Table 6
Long horizon task planning datasets.

Dataset Name	Description	Data Format	Number of Tasks	Task Descriptions	Dataset Size
TaPA [312]	Grounded planning with physical scene constraint	Image, text	-	Daily tasks	15k instructions
RoboCook [325]	Complex long-horizon soft body manipulation tasks	Point clouds	2	Complex long-horizon soft body manipulation tasks	19.73 GB
RLBench [86]	An ambitious large-scale benchmark and learning environment	Image, depth, joint angles, velocities and forces	100	Manipulation tasks	-
PackIt [361]	A dataset of hard packing problems	Point cloud	-	The geometry of objects and plan actions for manipulating	7.7 GB
SheetCopilot [362]	A dataset as a foundation for assessing the spreadsheet control capabilities	Text	221	Spreadsheet control tasks	-
Kinetics [363]	A dataset for inverse kinematics of robotic manipulators	Coordinates of the robotic manipulator end effector	-	Inverse kinematics	15k data
HMDB [364]	A large human motion database	Video	-	Human action tasks	2 GB
UCF [365]	An action recognition data set of realistic action videos	Video	101	Action recognition tasks	6.5 GB
CALVIN [366]	A benchmark to learn long-horizon language-conditioned tasks	Image, depth, tactile image	34	Language-conditioned continuous control tasks	1.3 GB
IKEA assembly dataset [367]	A multi-modal and multi-view video dataset of assembly tasks	3 RGB views, one depth stream, atomic actions, human poses, object segments, object tracking, and extrinsic camera calibration	-	Assembly tasks	406 GB
Epic-Kitchens [368]	A large-scale dataset in first-person (egocentric) vision	Image, audio	700	Daily activities in the kitchen	1.1 TB
TUM Kitchen DataSet [369]	A dataset of everyday manipulation activities for motion tracking and action recognition	Image, motion capture data, RFID tag readings	-	Action recognition tasks	-
MPII Cooking Activities Dataset [370]	A dataset which distinguishes 65 fine-grained activities	Image, video	65	Cooking activities	141.7 GB
Coin-Dataset [371]	A large-scale dataset for comprehensive instruction video analysis	Video	180	Video analysis tasks	476 h video
Youcook2 [372]	A large task-oriented, instructional video dataset	Video	2000	Cooking tasks	35.1 GB
Kitchen(CMU) [373]	A dataset contains multimodal measures of the human activity of subjects performing the tasks involved in cooking and food preparation.	Video, audio, IMU	5	The human activity of subjects performing the tasks involved in cooking and food preparation	-
MMACT [374]	A large-scale dataset for cross modal learning on human action understanding	RGB, keypoints, acceleration, gyroscope, orientation, Wi-Fi, pressure	37	3 major groups: complex actions, simple action, desk actions	36764 instances
FLOBOT Perception [375]	A real-world dataset for robotic cleaning in public spaces	Image, depth, LiDAR	3(scenes)	Robotic cleaning tasks	12.5 GB
Fusion360GalleryDataset	A dataset contains rich 2D and 3D geometry data derived from parametric CAD models	B-Rep, mesh, and point cloud	-	Design tasks	26.7 GB

logical reasoning, commonsense reasoning, affordance reasoning, and personalized reasoning [377–381].

Considering the autonomy and interactivity of robots, we classify reasoning. Logical reasoning focuses on strict regularity and certainty (autonomous completion requires formal logic), common sense reasoning emphasizes practicality and flexibility (autonomous completion requires daily experience), affordance reasoning emphasizes operability and interactivity (interaction with the environment and objects), and personalized reasoning focuses on individual uniqueness (interaction with people).

3.4.2. Methods

3.4.2.1. Logical Reasoning . Machine logical reasoning refers to the ability of a machine to draw new conclusions through the derivation of known facts and reasoning rules. In logical reasoning, the machine needs to take into account the logical relationship between facts, such as "if A is true, then B is also true" and "One of A and B must be chosen." In

addition, machines need to be able to understand fuzzy information, take into account the possibilities of different situations, and choose the most reasonable conclusion from them [382]. Chain-of-thoughts can significantly improve the complex reasoning capabilities of large language models [383]. Tree-of-thoughts can be used for tasks that require exploration, strategic foresight or where initial decision-making plays a key role [384].

Some studies have attempted to address the challenge of logical reasoning by using LLM generation. Liang et al. proposed a method to generate policies based on LLMs. By inputting natural language instructions, the LLM trained by code completion can write robot policy codes. These policies are reactive policies and waypoint-based policies. Features include exhibiting spatial-geometric reasoning, generalizing new instructions, and assigning exact values to ambiguous descriptions based on context [347].

Some studies use vision-language-action models to perform reasoning. To improve generalization and enable emergent semantic reasoning, RT-2 studies the integration of visual language models

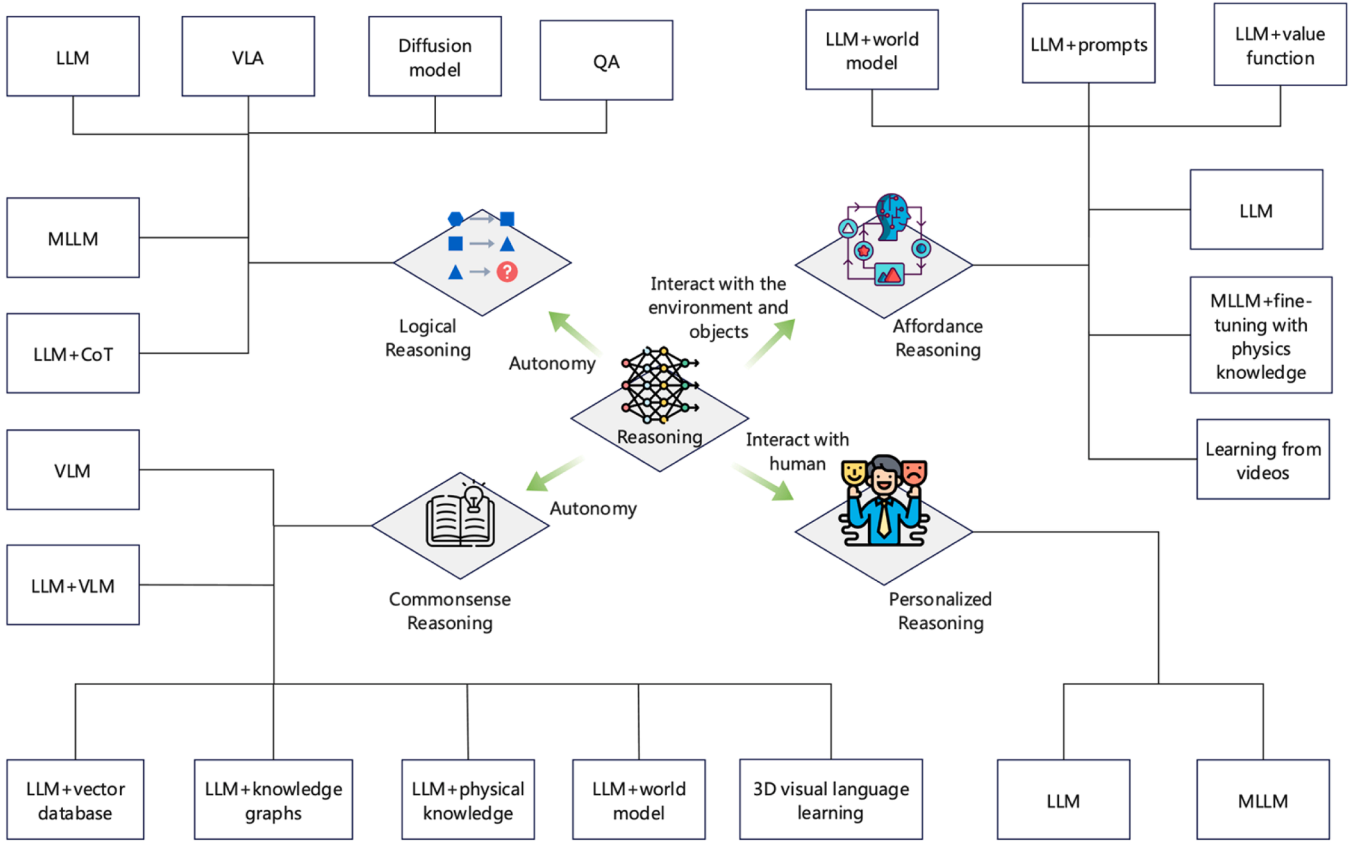


Fig. 7. Overall structure of reasoning.

trained on web data into end-to-end robot control. The authors propose to fine-tune the visual language model together with robot trajectory data to output robot actions. These actions are represented by textual tokens. This approach can achieve efficient robot policies and enables RT-2 to gain emergent capabilities from training [141]. RT-H uses language actions to build an action hierarchy, expanding the execution capabilities of the task [385].

Some studies use multimodal models or the integration of multimodal models with other models to solve reasoning problems. To address the challenge of enabling robots to achieve general reasoning in the real world, PaLM-E incorporates sensor modalities into the language model to establish a connection between words and perceptions [139]. The authors evaluate that PaLM-E is a single large-scale embodied multimodal model that can solve embodied reasoning tasks in a variety of implementations. Existing large-scale multimodal language models for robots still face two challenges: one is that the reasoning ability is not enough to handle complex tasks, and the second is that the computational cost of fine-tuning and reasoning is high. To address this problem, RoboMamba was proposed, which is an end-to-end large-scale multimodal language model for robots. It uses the Mamba model to provide robot reasoning and action capabilities while maintaining efficient fine-tuning and reasoning. In order to further enable RoboMamba to have the ability to predict action postures, the authors also explored an efficient fine-tuning strategy with a simple policy head [386].

The research uses network-level diffusion models for robotics. DALL-E-Bot [387] brings web-scale diffusion models to robotics. This method enables the robot to rearrange objects in the scene. Zero-shot inference is achieved using DALL-E-Bot without requiring any further data collection or training.

Some studies are based on the expansion of chain-of-thoughts for step-by-step reasoning. To solve various language reasoning and decision-making tasks, React proposes a general paradigm that

combines reasoning and action with language models. With the support of chain-of-thoughts prompting and external knowledge bases, LLMs generate reasoning trajectories and task-specific actions in an interactive manner, thereby exerting greater synergy between the two [388]. To mitigate reasoning fallacies and draw inspiration from multi-agent collaboration, the CaCo-CoT framework is proposed to improve the fidelity and causality of knowledge-based reasoning. The framework consists of two types of agents (reasoners and evaluators) that work together in a reasoning and consensus paradigm to ensure causal consistency [389].

Some studies are based on question-answering methods for reasoning. To solve the knowledge-based embodied question-answering task, Tan et al. proposed a new framework based on neural program synthesis reasoning, in which external knowledge and 3D scene graphs are jointly reasoned to achieve navigation and question answering [390].

There are still some valuable methods for reasoning that are worth studying, including accumulative reasoning [391], context learning [392] and causal reinforcement learning [393].

3.4.2.2. Commonsense Reasoning. LLMs are black-box models that typically fail to acquire and capture factual knowledge. In contrast, knowledge graphs, etc., are knowledge models that explicitly store rich factual knowledge. Therefore, they can complement each other's strengths and play a better role in reasoning.

LLMs combined with knowledge graphs is a way to try common sense reasoning. Knowledge graphs can enhance LLMs by providing external knowledge for reasoning and interpretability. At the same time, knowledge graphs are inherently difficult to construct and evolve, posing challenges to existing methods for generating new knowledge and representing unseen knowledge in knowledge graphs. Therefore, it is complementary to combine LLMs and knowledge graphs to play their

respective advantages at the same time [394]. LLMs and knowledge graphs (KGs) can complement each other: LLMs can be used for KG construction or completion, while existing KGs can be used for different tasks, such as making LLMs' outputs interpretable or fact-checking in a neural-symbolic manner. Text2KGBench is a benchmark that evaluates the ability of language models to generate knowledge graphs from ontology-guided natural language text [395]. Kim et al. proposed KG-GPT, a general framework that utilizes LLM for knowledge graph reasoning [396]. Zhu et al. proposed AutoKG, a method based on multi-agents, using LLM for knowledge graph construction and reasoning [397]. Yang et al. proposed to enhance the large language model KGLLM by developing knowledge graphs and provided a solution to enhance the factual reasoning ability of LLMs [398]. Tang et al. proposed GraphGPT, which utilizes LLMs for graph instruction tuning [399]. In order to mitigate the operational risks of service robots, Qi et al. proposed combining embodied robot control prompts (ERCP), large language models (LLM), and embodied knowledge graphs (EKG) to ensure that human-robot interaction is safety-oriented [400]. FireRobKG is a knowledge graph developed for fire prevention. Its construction process can also be applied to other robotics scenarios. FireRobBrain uses a prompt module to combine FireRobKG with LLM to assist firefighting robot planning [401].

Some studies are integrating large language models with physical knowledge to explore commonsense reasoning. Zellers et al. proposed QLeT, a model that learns physical commonsense knowledge through interaction and then uses this knowledge to construct language. The authors decompose QLeT into a physical dynamics model and a separate language model. Using a dynamics model as an interface to a language model, QLeT can read a sentence, neurally simulate what might happen next, and then communicate that result through a text-symbolic representation or natural language [402].

Some studies combine large language models with world models to overcome the reasoning problem. Surfer is a framework for robot manipulation using world models. Based on the world model, it decouples robot manipulation into two parts: action and scene. By modeling action scene prediction, the model's generalization ability to new instructions and new scenes is enhanced. The authors also created SeaWave, a robot manipulation benchmark with progressive reasoning tasks [403].

Some studies have combined large language models with visual language models to explore commonsense reasoning. Kwon et al. used large language models and visual language models to help robots actively perceive their environment to perform solid commonsense reasoning [404].

Some research has applied visual language models to commonsense reasoning. The VLM-PC system uses visual language models to perform commonsense reasoning on legged robot adaptation. VLM enables robots to perceive, navigate, and act autonomously in a variety of complex scenarios that would otherwise require environment-specific engineering or human guidance [405].

Some research works try to use 3D visual language learning for contextual reasoning. 3D-VisTA is a pre-trained transformer for 3D vision and text alignment. The model uses only self-attention layers and can be easily adapted to downstream tasks without any auxiliary losses or optimization tricks. The project team also released the ScanScribe dataset, a 3D scene-text pairs dataset for 3D-VL pre-training [98].

One approach to address inference problems is by incorporating vector databases. Both LlamaIndex and Langchain are working diligently to develop a data-augmented retrieval system, which could be further enhanced with a contextual agent. Yohei introduces the concept of incorporating relevant contextual information (task context), which may have nuances differing from those of the conventional semantic similarity algorithm offered by vector databases. In addition, RAG can alleviate the hallucination problem of large language models to a certain extent. That is, it provides domain-specific data with accurate instructions to large language models. The vector database stores domain-

specific data as an external knowledge base so that large language models can easily process a large amount of data owned by users [406].

There are also some other studies worth paying attention to. Ren et al. propose KNOWNO, a framework for measuring and adjusting uncertainty in LLM planners so that they know when they don't know and seek help when needed. KNOWNO builds on conformal prediction theory to provide statistical guarantees for task completion. At the same time, human assistance in complex multi-step planning setups is minimized [407].

3.4.2.3. Affordance Reasoning. The ability to reason about affordances enables robots to choose actions that are appropriate for a given object and produce a desired effect.

Some studies combine large language models with world models to improve the robot's affordance reasoning ability. The Dreamer algorithm has recently shown great promise for learning from a small number of interactions by planning in a learned world model, outperforming pure reinforcement learning in video games. Learning world models to predict the outcomes of potential actions can plan in imagination, reducing the amount of trial and error required in real settings [408]. Yoneda et al. published the Statler framework, which enables LLMs to have representations of world states that change over time while maintaining memory. The framework has two generic LLM instances: a world model reader and a world model writer. These two parts interact with and maintain the world state. With access to this world-state memory, the Statler framework improves the ability of existing LLMs to reason over longer periods of time, independent of context length constraints. Experiments on simulated domains and real robots show that the proposed method improves the state-of-the-art in LLM-based robot inference [409].

Some studies have used large language models and prompts to improve affordance reasoning capabilities. Robots need primary knowledge of the world in which to act. LLMs can be used to score potential subsequent actions during task planning or even directly generate action sequences without natural language instructions from additional domain information. Singh et al. propose a procedural LLM hint structure that enables plan generation to work across environments, robot functions, and tasks. LLMs are prompted by program-like specifications of operations and objects available in the environment, as well as example programs that can be executed. Situational awareness is introduced in LLM-based robot task planning [410]. Tang et al. proposed a knowledge condition detection framework, CoTDet, for affordance knowledge prompts for task-driven object detection [411].

Some research improves robots' affordance reasoning capabilities by combining large language models with value functions for training skills. SayCan incorporates real-world experience into LLMs through the value function of pre-trained skills, enabling LLMs to execute real-world abstract, long-horizon commands on robots. This method implements language models to provide high-level semantic knowledge and provide pre-trained low-level skills to constrain the model to propose natural language actions that are both feasible and appropriate to the context [412].

Some studies have improved affordance reasoning capabilities by creating physical knowledge datasets and using them to fine-tune large multimodal models. Gao et al. fine-tuned VLM on PhysObjects to improve its understanding of physical object concepts by capturing human priors on these concepts from visual appearance. Combining this physics-based VLM with a large-language-model-based robot planner into an interactive framework improves mission success rates [413]. ManipVQA injects robot affordances and physical grounding information into large multimodal models to improve efficiency in manipulation tasks [244].

Some studies have improved affordance reasoning capabilities by interacting with LLMs. RAIL [414] enables robotic affordance imagination via large language models. The system adopts a hierarchical

framework consisting of analysis, imagination, and evaluation to perform novel tasks on unseen objects based on affordance reasoning.

Some research has used human videos to learn visual affordances. VRB [415] uses human videos to learn visual affordances. The method trains a visual affordance model that estimates where and how humans are likely to interact in a scene. These affordances can be deployed on multiple real robots. And the structure of these behavioral affordances directly enables robots to perform many complex tasks.

In addition, to study how humans interact with objects, some works have studied affordance datasets [416,417].

3.4.2.4. Personalized Reasoning. The emergence of large language models marks a revolutionary breakthrough in artificial intelligence. A major leap forward in the capabilities of general artificial intelligence will change how personalization is implemented. On the one hand, it will change the way humans interact with personalized systems. On the other hand, it will also greatly expand the scope of personalization [418].

Some research explores personalized reasoning capabilities in large

language models. The growing use of LLMs in conversational agents has sparked interest in the personality exhibited by data-trained models, as personality significantly influences communication effectiveness. Therefore, Safdari et al. proposed a comprehensive approach to test the personality traits expressed in the text generated by LLMs. Experiments find the reliability and validity of LLMs for simulating personality for larger and fine-tuned models. The personalities in the LLM output can be shaped in desired dimensions to mimic specific personality traits [419]. Huang et al. proposed to assess the empathic ability of LLMs, that is, how their feelings change when they encounter a specific situation. After experimental evaluation, LLMs can usually respond appropriately to some situations, although there are some biases. Still, they don't match human emotional behavior to make connections between similar situations. This paper expects to contribute to the advancement of LLMs to better adapt to human emotional behavior, thereby enhancing the practicality and applicability of intelligent assistants [420]. Wu et al. proposed TidyBot. This method can be used for robots to learn personal preferences and perform personalized cleaning of the room. The robot uses the planning and perception of language combined with the

Table 7
Reasoning datasets.

Dataset Name	Description	Domain/Application	Data Type	Task Type	Size (Number of Samples)	Publisher
InfLevel [424]	A benchmark designed to evaluate AI systems' understanding of core physical principles	Evaluate AI systems' understanding of core physical principles	Video	Manipulation	80k videos	AI2
Socratis [422]	A benchmark of diverse open-ended emotional reactions to image-caption pairs	Predict human emotion at an image and caption pair	Image, text	Reasoning	18378 annotations	Boston University
RoboTurk [81]	A crowdsourced teleoperated robot manipulation dataset	Challenging manipulation tasks	Video	Manipulation	2144 demonstrations	Stanford University
Rainbow [425]	A universal commonsense reasoning benchmark spanning both social and physical common sense	Commonsense reasoning tasks	Question-answering	Reasoning	200+ experiments	AI2
PhysObjects [413]	An object-centric dataset of common household objects	Help VLMs understand the physical concepts (e.g., material, fragility) of common objects	Image, text	Manipulation	39.6k crowd-sourced and 417k annotations	Stanford University, etc.
AGENT [426]	A benchmark for core psychological reasoning	Core psychological reasoning	RGB-D frames, segmentation maps, GT states	Reasoning	3360 trials	MIT, etc.
Winograd Schema Challenge [427]	A dataset for commonsense reasoning and AI.	Commonsense reasoning	Text	Multiple Choice Questions	150 schemas	University of Toronto
SWAG [428] (Situation With Adversarial Generations)	A dataset for grounded commonsense inference.	Commonsense reasoning	Text	Multiple Choice Questions	113k examples	University of Washington
CLEVR (Compositional Language and Elementary Visual Reasoning) [429]	A dataset for visual reasoning and logic.	Visual reasoning	Images, text	Multiple Choice Questions	100k images, 100k questions	Stanford University
DROP [430]	A dataset for reading comprehension and reasoning.	Reading comprehension	Text	Question Answering	96k questions	AI2
SQuAD (The Stanford Question Answering Dataset) [431]	A reading comprehension dataset	Reading comprehension	Text	Question Answering	150k questions	Stanford University
ARC [432] (AI2 Reasoning Challenge)	A set of science questions that require various forms of reasoning.	Science reasoning	Text	Multiple Choice Questions	7k questions	AI2
HellaSwag [433]	A dataset designed for commonsense reasoning with unconventional questions.	Commonsense reasoning	Text	Multiple Choice Questions	70k questions	AI2
QuAC [434] (Question Answering in Context)	A dataset for reasoning about questions posed in a dialogue context.	Dialog-based reasoning	Text	Question Answering	14k dialogues	University of Washington
OBQA [435] (OpenBookQA)	A dataset for open-domain question answering that requires reasoning over a large collection of texts.	Question answering	Text	Multiple Choice Questions	6k questions	AI2 and University of Washington

summarization ability of LLM to pick up objects, determine the placement of objects, and tidy the room [166]. LLM-Personalize [421] aligns LLM planning with human preferences through reinforced self-training of domestic robots.

Some studies explore personalized reasoning capabilities in large multimodal models. Deng et al. present Socratis, a social response benchmark that tests the ability of state-of-the-art multimodal large language models to generate emotional reasons for a given image-caption pair [422]. EMER can achieve reliable and open-vocabulary emotion recognition. At the same time, EMER can also serve as a basic task to evaluate the audio-video-text understanding ability of MLLM. To promote the research of EMER, the authors established an initial dataset, developed a baseline, and defined evaluation metrics [423].

3.4.3. Datasets and metrics

Datasets. Robot reasoning datasets typically involve tasks that require problem-solving abilities such as logical reasoning and common-sense reasoning. These datasets are designed to evaluate the robot's ability to make inferences in various scenarios. Here are some examples of robot reasoning datasets in Table 7.

Metrics. The evaluation indicators for robot reasoning datasets typically depend on the specific dataset and task. However, common evaluation metrics and indicators are used for many reasoning datasets.

Reasoning performance is evaluated by (1) Accuracy. (2) F1 Score: The F1 score is calculated based on the overlap between predicted answers and ground truth answers. (3) Exact Match (EM): EM calculates the percentage of questions answered exactly correctly. (4) Top-k Accuracy. (5) Mean Reciprocal Rank (MRR): MRR measures the quality of the top-ranked answer. It calculates the average reciprocal rank of the first correct answer in the ranked list of answers. (6) BLEU (Bilingual Evaluation Understudy): Commonly used in machine translation tasks, BLEU measures the similarity between predicted and reference answers.

3.4.4. Problems

Robot learning goals are to master how to learn, combine advanced pattern recognition with model-based reasoning, and develop common sense intelligence. With the advancement of learning and the improvement of intelligence [436], research on robot reasoning has gradually deepened. Reasoning is an abstract, advanced form of thinking. The objective basis of reasoning is the relationship between objective things. At present, the reasoning capabilities of robots are weak (including causal reasoning [437], spatiotemporal reasoning, real-time reasoning, geometric reasoning, world model, world knowledge, common sense reasoning, understanding of physical constraints, etc.). One very important point is that the robot lacks reverse reasoning, and the basic model may reason from left to right. The current model is based on text-visual speech, so it does not directly reason about information such as touch, and more powerful multi-modal models are needed when available.

Solutions that can be tried are as follows: Beyond statistical correlation, reasoning about system dynamics and causality; meta-learning with limited data; rapid learning to adapt to dynamic, uncertain environments; learning across heterogeneous tasks and domains; Developing systems that know their limitations and know how to ask for help; Developing systems that can deeply understand and synthesize complex textual and narrative information; Conducting deep moral and social reasoning about real-world problems.

4. Discussion

Despite the significant progress made in robot learning based on foundation models, the field still faces numerous challenges in terms of both technical aspects and ethical aspects. In the following section, we will outline the major challenges, potential solutions, and potential future directions. We hope that the highlighted aspects can serve as inspiration for future research investigations in the robot learning area.

4.1. Robot hardware and software decoupling

The basic requirement for a foundation model is a unified architectural framework. However, there is significant variation in robot hardware, making it crucial to achieve bottom-level uniformity or decoupling. To facilitate the advancement of robot technology, it is crucial to ensure the synchronization of form with function [438], which implies that software and hardware must evolve simultaneously. Nevertheless, numerous challenges persist in the development of software and hardware. On the one hand, due to the diversity of manufacturers' specifications and private parameters, it becomes challenging to individually program and manage robots when (re)configuring them to achieve desired tasks or formations. This dilemma often occurs in industrial and operational fields. On the other hand, the construction of current robot models and databases largely depends on the hardware structure of the robots. The majority of robot databases are constructed by collecting specific data pertaining to individual robots. Consequently, trained robot models exhibit optimal performance solely on the particular robot they are trained for. Put simply, existing robot models are limited by the hardware structure of the robots and software algorithms.

Ideally, there is a need to decouple robot software and hardware and collect databases encompassing various robot models concurrently. The separation of the logical and physical components would foster increased software innovations and facilitate a potentially more accessible robot market, inevitably resulting in reduced non-recurring engineering costs. In a decoupled architecture, robot hardware and software can be developed and updated independently without restrictions. Additionally, the most recent high-performance software algorithms can be efficiently implemented across various robot models. Confronted with diverse hardware configurations and unpredictable operating environments, users can develop and program robots of different types without the need for a comprehensive understanding of the specific hardware employed by each robot. Compatible APIs, predefined libraries, and programming software can be developed and enhanced. The abstraction level of robot software needs to be improved to increase the efficiency and effectiveness of robot operations, enabling robot applications to run robustly in dynamic environments [439].

4.2. Dynamic data for interaction with the environment

When establishing a specialized large-scale robot model, it is necessary to utilize dynamic and diverse data [440], including dynamics data, during the training or fine-tuning process. Meanwhile, robots are anticipated to embody key attributes, including agility, cost-effectiveness, diversity, environmental adaptability, and plasticity, which empower them to execute tasks like fastening garments and tying shoelaces. Interactive environments [441] and abundant dynamic data are imperative for sufficiently training and assessing robots possessing these aforementioned attributes. However, the realization of authentic robot scenarios and the acquisition of real-time data present formidable challenges. Therefore, collaboration and resource sharing among laboratories worldwide assume paramount importance in propelling responsible and open advancements in robotics research. Data collection methods may encompass several approaches, including action library aggregation, teleoperation, and imitation learning. The crux of the challenge lies in dissociating the collected data from the specific robot model employed, thus ensuring its perpetual validity. Additionally, the existing training process in robot learning suffers from an inadequacy of effective and abundant data. Establishing multimodal databases that can be utilized for robot learning training is of paramount importance.

In order to better understand the environment, recognize objects, and perform tasks, current robot models often only focus on the fusion of top-level data, such as vision and language in multimodal fusion, while neglecting bottom-level data, such as dynamics data related to interaction with the environment. In real-world environments, there is no doubt that mechanics directly determine the stability of grasping. In

order to improve the accuracy, generalization, and robustness of robotic models and address issues related to force perception, it is necessary to construct large-scale dynamics data during the training process. Dynamics data includes information such as position (XYZ, rotation angles), acceleration, and forces (magnitude and torque). Based on this information, robots can easily manipulate objects of different materials and weights accurately. Understanding the robot's dynamics model allows the model to comprehend the robot from a bottom-up approach during the training process of existing robot learning. Incorporating dynamics data into the robot's foundation model also enables the model to generate more realistic and physically meaningful data.

4.3. Robot generalization

Robot models need to demonstrate better generalization ability, surpassing the semantic, visual, and other multimodal understanding of the robot data they encounter. It requires robots to perform operational tasks on objects or scenes in the robot data that they have never seen before. This necessitates the utilization of knowledge derived from network data for operation. Robot models should not only maintain performance on the original tasks in robot data but also improve performance in previously unseen scenarios. Accordingly, we present the following key elements.

Currently, the advancements in robotic research have yet to reach optimal levels of efficacy. A clear divergence is evident between human intelligence and that of robots, especially when evaluating areas such as image recognition and the competency of platforms like ChatGPT in question-answering scenarios. The enhancement of both the precision and efficiency of robotic learning stands as a critical imperative. While modern robotic learning methodologies have attained certain milestones, the scope for refining execution success rates remains substantial.

The environment is expected to become increasingly complex. Robots need to adapt to various scenarios to perform a variety of tasks. It is difficult to achieve long-horizon autonomy in complex environments, so lifelong learning is required. Currently, robots are only capable of performing a limited number of skilled tasks and are not generalizable across a variety of tasks [442]. On the one hand, Robots will adapt to different scenarios, including layout configuration, visual texture, light source changes, time simulation, low cost, data enhancement, demand personalization, and program automation. In virtual-real transfer, methods that may be used include domain adaptation, meta-learning, transfer learning, knowledge distillation, and world models. Legged robots have made a lot of progress in traditional methods, but they lack the ability to combine with foundation models. On the other hand, Robots can observe, understand, and practice in interaction with their environment, thus self-exam, diagnose, and repair. The robot can not only collect data by itself but also analyze the data, analyze the cause of the failure, and then solve the problem. Robots can have self-driven learning awareness and can continue to learn [358].

Currently, foundation models are developed and trained in unmanned environments. In the future, when they are utilized in environments with human presence, interactions between robots and humans will need to take human factors into consideration. It requires human-robot alignment and considers emotional factors. It also considers biomimetic learning, cross-learning, and the co-development of the brain and the body.

AI alignment is worthy of in-depth research. The goals of AI systems are required to be consistent with human values and interests. If the values of AI and humans cannot be aligned, the following problems will arise: behaviors that do not meet the goals of human intentions, loss of control, harm to human interests, and incorrect choices in the conflict of multiple setting goals [331,443].

Researchers can combine functionality with expressive capabilities and explore methods to improve robots from this perspective. Functionality is manifested in mobility, dexterity, perception, and

intelligence. Considering the relationship between individuals and occupations, we can liberate people from dangerous situations and enhance the technological level of humanity.

While current foundation models are capable of engaging in high-level semantic conversations, they fall short of humans in low-level control aspects such as movement and operation. Consequently, bionics is deemed significant. Biomimetic learning is a highly worthwhile field of research that is used to design and develop more powerful robots. It draws inspiration from biological systems, especially animals and organisms. By observing how animals and organisms move, perceive their environment, and interact with it, researchers attempt to replicate these mechanisms in robots. They utilize biomimetic materials, such as artificial muscles and soft robot components, to enable robots to imitate the adaptability and flexibility of living organisms. The objective of researchers is not only to replicate the physical characteristics of animals but also to understand their behavioral patterns. This includes understanding how animals make decisions, solve problems, and exhibit intelligent behavior. Overall, the aim of biomimetic learning in robotics is to push the limits of robot capabilities by harnessing natural wisdom and integrating biological principles into the design and control of robots, making them more versatile and adaptable to various tasks and environments.

Robots can achieve dexterous movement and manipulation. Research has found that the perception-action circuit is the center of cognition, and the body uses the perception-motor system to generate intelligence in the interaction with the environment. Intelligent robots span multiple directions in intelligent disciplines, such as cognitive science, psychology, brain science, and sociology. Therefore, the focus is on interdisciplinary integration, analogous to the developmental process of human intelligence. In multi-modal environmental interaction, by opening up the links and loops between morphology, perception, behavior, and learning, robots can realize active perception and autonomous learning.

Being conscious is hard, but the body and brain of a robot should develop together [438,444]. In robot learning, the robot lacks common sense and lacks the physical constraints of learning real scenes from the environment. Robot learning is to make robots intelligent. In addition to perceptual and motor capabilities, which have been studied in the past, cognitive abilities are also very significant. Referring to the development process of human intelligence, the brain and body develop simultaneously. Intelligent robots not only have perception and motor intelligence but also cognitive intelligence with autonomous consciousness. In addition, robots lack the ability to learn actively. For example, when completing the specified target tasks, it can actively ask what it needs to do next in order to achieve a certain task. Active learning may regain favor as autonomous contextual agents actively reveal what they don't know in order to prioritize [445].

4.4. Multimodal interaction

Currently, the interaction data of robots mainly relies on visual and textual information, lacking diversity. In order to enhance the robot's perception of the environment and its interaction with it, it is necessary to integrate multimodal data from various sensory modalities, including visual, auditory, tactile, olfactory, gustatory, and other sensory inputs. Although multimodal data brings great hopes for advancing robot technology, it also comes with some problems and challenges. Specifically, different modalities have different data formats, structures, and features, posing challenges for the collaborative interaction among different modalities. The main issues involved are multimodal representation, multimodal mapping, multimodal alignment, and multimodal fusion.

To address these problems, several considerations can be taken into account. Research can be conducted to tackle challenges related to large-scale environments, multitasking, and strong interaction. Tasks with higher interactivity can be verified and demonstrated in more open and

complex environments, such as simulators and multi-platform settings. Various fusion strategies can be explored to effectively combine information from different modalities, including hierarchical and multilingual fusion, as well as scalable fusion. Advanced deep learning models, such as multimodal transformers, can be developed to efficiently fuse and extract meaningful information from different sensory modalities. Cross-modal learning techniques can be explored to enable robots to learn meaningful correlations between different modalities, thereby improving their understanding of the environment. Transparent and interpretable multimodal fusion models can be developed to facilitate human understanding and trust in robot decisions. The capability of robots to understand the semantics of multimodal data, including scene understanding [17] and context comprehension, can also be enhanced. Additionally, during the process of converting multimodal content, foundation models may lose information or generate errors, leading to biased results. This requires foundation models to achieve strong multimodal understanding, addressing the current issues of unreliability in intelligent agents, as well as information loss and biases.

4.5. Robotics-specific foundation models

Foundation models have achieved many gratifying results in text, language, etc., but in the field of robotics, the development of foundation models is slow. There is an urgent need to develop usable and effective general-purpose foundation models and special-purpose foundation models for robots. In order for robots to reach human-like levels of capability, it is necessary to collect robot data for every object, environment, task, and situation. In addition to the issue of robot data, the following problems and challenges also need to be overcome.

Multimodal large language models suffer from the catastrophic forgetting problem. Fine-tuning a model can improve specific performance, but as fine-tuning proceeds, the model begins to exhibit hallucinations, resulting in a significant loss of generality [446]. The hallucination problem of foundation models is that the text generated by the model does not follow the original text (Faithfulness) or does not conform to the facts (Factualness) [447]. As a result, the robot gives wrong judgments or operations in the process of performing tasks in combination with the foundation model [448]. Currently, in order to obtain reasonable answers in multi-turn dialogues with foundation models, multiple prompts need to be provided. Therefore, it is necessary to design foundation models that do not rely on human prompts. Furthermore, foundation models need to have better contextual understanding and causal understanding [449,450].

Combining large models and small models is also a method that can be tried [451]. Large models are well suited to handle Corner Cases, while small models trained by modeling and for scenarios cannot exhaustively cover the whole scenario, and this part of the work can be handled by the general understanding and strong generalization and inference capabilities of large models. The combination of large models and small models can improve the inference efficiency without reducing the inference ability.

Foundation models for low-level control need to be designed. Due to the lack of data for low-level controllers in the training corpora, most of the existing robot foundation models have been applied to robots as either a semantic planner or to interact with robots using human-designed action primitives [452]. Having a large embodied model enables one-step control of the lowest level.

The evaluation of embodied intelligent agents should shift from task-oriented evaluation to capability-value evaluation. Human discrimination tests evaluate AI based on human observations, as represented by the classic Turing test. Qualitative testing can only be done through human observation and there is no quantitative testing [453–459].

While LLMs have been utilized in downstream tasks requiring task planning, various non-language-related tasks in robotics, such as fetching eggs and pouring water, also benefit from foundation models of various modalities. The integration of visual and linguistic modalities,

for instance, has demonstrated significant potential for self-supervised learning, which is crucial for navigation. In the future, multi-modal large models, particularly those that incorporate dynamic fusion, will play a vital role in manipulation tasks. Moreover, large embodied models capable of continuously interacting with, perceiving, and understanding the environment, as well as synthesizing multi-modal information, are essential for complex tasks involving robot planning and reasoning.

In terms of integrating data-driven methods, we explore two primary strategies. The first strategy involves employing a pre-trained foundation model and adapting it to downstream tasks through prompt engineering or fine-tuning. The second strategy entails gathering robotic data to train a dedicated foundation model for the robot.

4.6. Computational efficiency

Computational efficiency [460] requires continuous evolution and improvement. Optimization aspects while ensuring high performance mainly include the following three aspects: computing costs need to be reduced, computing resources need to be reduced, and computing time needs to be reduced.

There has been a staggering increase in computational power requirements. However, storage performance significantly falls behind that of processors. These two factors contribute to the problems of the computational wall and the storage wall. In response to the above challenges, researchers are committed to addressing issues related to novel AI storage and computing technologies. These technologies aim to break through the storage and computational bottlenecks of AI calculations, improve computational efficiency, and mainly include new applications, computing frameworks, storage-computing architectures, and cloud infrastructure technologies. In terms of hardware architecture, there are two aspects: von Neumann architecture and novel storage-computing architecture. The novel storage-computing architecture can focus on integrated storage-computing chips, neuromorphic chips, and so on.

The models involved in the robot system require efficient computing, including the following aspects: cloud-edge-device integration and smart chips. Cloud-edge-device integration requires unified management, cloud-edge collaboration, and resource allocation. Smart chips need to be small in size, low in power consumption, and high in performance.

The existing foundation models have a huge amount of parameters, and the training of parameters requires the support of many high-performance graphics cards, which requires a huge cost. In order to reduce the cost and the amount of parameters of the model, a lightweight foundation model is designed while ensuring the performance [461]. Additionally, further compression can be performed on the network structure of large-scale models. In addition to adjusting the parameter size, extraction methods such as knowledge distillation, network pruning, low-rank parameter decomposition, and quantization can also be employed.

4.7. Robot safety and ethics

Robot security includes two aspects: physical security and data security. Physical safety refers to the fact that some misunderstandings when combining foundation models with robots may lead to unexpected chain reactions. For example, the robot received instructions to cook Western food in the oven, but it cooked Chinese food. As a result, it turned on the gas and accidentally caught fire. The specificity of robot tasks lies in the constant interaction with the environment during task execution, and robot safety becomes particularly crucial when humans are present in the environment. In the process of completing the specified target tasks, the robot lacks a safety guarantee due to physical constraints and other real environmental restrictions, and there may be accidents such as collisions, extrusions, and damage to mechanical parts,

potentially causing harm to humans.

Data security is equally important, and it involves data privacy. Privacy risks in robot foundation model development and application primarily come from the information contained in the original training data and the powerful inference capabilities of the models. Developers need to ensure that robot foundation models do not cause privacy breaches and carefully evaluate the potential ethical issues they may bring about. In addition to the data bias in the foundation models being trained, every user will also have information security concerns when training the foundation model and uploading the data to the cloud. The intellectual property rights and data security cannot be guaranteed. This requires the use of data desensitization. In addition to risks such as sensitive information leakage and content infringement, there are also risks such as model denial of service, model theft, training data poisoning, model hallucination, and attacker prompt injection. Therefore, it is necessary to establish a comprehensive and multi-faceted security evaluation, detection, and defense system.

Ethically, the behavior of robots should comply with social and legal norms. Developing autonomous robots with self-awareness is also a promising direction. There is growing concern about the ethical and safety aspects of robot learning, including fairness, transparency, and robustness. It is important to pay attention to biases and toxicity in the training samples of robot foundation models to ensure that robots' behavior does not lead to discrimination or unfairness.

5. Conclusion

This paper provides an overview of the main challenges in robot learning. It also provides an inductive and comparative analysis of the types of algorithms that have been developed to address these challenges by combining robot learning with foundation models. We outlined the development and evolution of related technologies for robot learning, as well as the prerequisites such as datasets and computing resources required. We divided these key robot learning challenges into four categories according to downstream tasks, namely manipulation, navigation, task planning, and reasoning. With the development of foundation models, these models have demonstrated significant progress in robotic applications and promising humanoid intelligence [462]. These findings present a bright future for foundational models in robot applications. Last but not least, discussions were conducted, which explained the current problems and challenges of robot learning and proposed research directions in the future, including robot hardware and software decoupling, dynamic data for interaction with the environment, robotics-specific foundation models, among others.

CRedit authorship contribution statement

Xuan Xiao: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Jiahang Liu:** Investigation, Formal analysis, Data curation. **Zhipeng Wang:** Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Yanmin Zhou:** Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Yong Qi:** Writing – review & editing, Resources, Investigation. **Shuo Jiang:** Writing – review & editing, Visualization, Supervision, Resources, Investigation, Funding acquisition, Conceptualization. **Bin He:** Writing – review & editing, Resources, Funding acquisition. **Qian Cheng:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Compliance with ethics guidelines

Xuan Xiao, Jiahang Liu, Zhipeng Wang, Yanmin Zhou, Yong Qi, Shuo Jiang, Bin He, and Qian Cheng declare that they have no conflict of interest or financial conflicts to disclose.

Data availability

Data will be made available on request.

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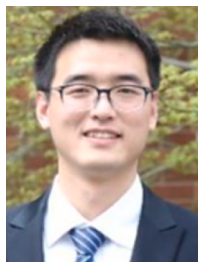
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